

Coordination of turn-related activity in the superior colliculus with locomotor dynamics and hippocampal representations of possible futures

Cameron Wilhite¹, Loren M. Frank^{2,3,4,5,6}, Massimo Scanziani^{2,4,5,6,7}

¹Neuroscience Graduate Program, University of California, San Francisco, CA, USA

²Department of Physiology, University of California, San Francisco, CA, USA

³Department of Psychiatry, University of California, San Francisco, CA, USA

⁴Howard Hughes Medical Institute, Chevy Chase, MD, USA

⁵Kavli Institute for Fundamental Neuroscience, University of California, San Francisco, CA, USA.

⁶Senior author

⁷Lead contact

Correspondence: Cameron Wilhite, camwilhite@gmail.com, Loren M. Frank, Loren.Frank@ucsf.edu, Massimo Scanziani, massimo@ucsf.edu

SUMMARY

Turning left or right is a core element of animal navigation. Such turns occur in precise coordination with the ongoing stepping rhythm and require internal planning for the animal's future path. The superior colliculus (SC) is a midbrain structure involved in turning. Whether turn-related activity in the SC is coordinated with stepping and internal representations of future paths is unknown. Here, while recording from left- or right-preferring “turn cells” in the motor layers of the SC as mice navigated a Y-maze, we monitored locomotor dynamics and decoded internal representations of future paths from the hippocampus. We discovered that turn cell activity was tightly phase-locked to the stepping rhythm and modulated in coordination with hippocampal representations of future paths. The coordination of turn-related activity in the SC with stepping and internal representations of future paths may allow animals to seamlessly execute turns during locomotion while navigating toward planned destinations.

INTRODUCTION

An animal's path through its environment is shaped by a series of left and right turns. These changes in locomotion direction are fundamental components of spatial navigation and goal-directed behavior. While such turns unfold seamlessly along the animal's trajectory, they are in fact precisely coordinated with the ongoing locomotor stepping rhythm and require internal planning for the animal's future path. On one hand, coordinating turns with specific phases of the stepping cycle optimizes efficiency and stability during turning¹⁻³. On the other hand, internal planning for the animal's future path ensures smooth and accurate trajectories during turns^{4,5}. Whether the activity of brain structures directly involved in turning is coordinated with both the stepping rhythm and internal representations of future paths is unknown.

Here, we tested this possibility in the superior colliculus (SC), a conserved sensorimotor midbrain structure that plays a central role in orienting movements^{6,7}, like left and right turns. Studies in rodents have shown that the intermediate and deep motor layers of the SC (dSC) contain neurons whose activity reports turn direction⁸⁻¹² and whose stimulation triggers turning movements^{13,14}. However, neuronal activity in the dSC has been primarily studied in movement-restricted animals performing turns in response to external stimuli^{8,9}. Accordingly, little is known about how turn-related activity in dSC unfolds during freely-moving, internally-generated behaviors, like spatial navigation¹⁵, when turns need to be coordinated with both stepping and internal representations of future paths.

Is turn-related activity in the dSC modulated in coordination with the animal's ongoing stepping rhythm and with internal representations of future paths? To address this question, we recorded from neurons in the dSC in navigating mice while tracking locomotor dynamics with a camera and decoding internal representations of future paths from hippocampal place cell activity¹⁶⁻¹⁸.

We discover that the activity of left- or right-preferring 'turn cells' in the dSC is tightly phase-locked to the ongoing locomotor stepping rhythm such that left and right turn cells fire at opposite phases of the stepping cycle. Notably, this alternation is already present during straight locomotion, before the animal initiates the turn. Furthermore, we reveal that turn cell activity is modulated in coordination with hippocampal representations of future paths. Specifically, hippocampal representations of possible left or right future paths predict increased firing rates of dSC neurons with matching turn preference. These results reveal how turn-related activity in the SC, perhaps one of the best-understood motor outputs of this structure, is highly integrated with the ongoing behavior and internal plans of the organism as it navigates through its world.

RESULTS

We used silicon probes to record from dSC neurons (Figure 1A, top, and Figure S1) as mice performed a Y-maze spatial navigation task (Figure 1A, middle). Each task session consisted of multiple trial blocks during which two of the three reward ports (located at the ends of the maze arms) were active, with their locations changing uncued across blocks. This trial block design was chosen to maintain cognitive engagement with the task. During each trial, the mouse navigated from one reward port to another and, across trials, consistently alternated between the active ports, resulting in repeated left and right turns at the maze bifurcation. We first identified neurons whose activity reported turn direction around the bifurcation ('turn cells'). We then determined whether their activity was modulated in coordination with the animal's stepping rhythm and with hippocampal representations of future paths.

Turn cells in the dSC

Turn cells were defined as neurons that showed significant firing rate modulation around the bifurcation of the maze (significant Turn Direction Selectivity Index, or TDSI, see Methods), consistently in the same direction, regardless of which maze arm the animal came from (Figure 1B and 1C). Turn cells accounted for ~30% of all dSC cells (28%, 411/1471, $n = 12$ animals). Turn cell firing became directionally selective ~300 ms before the bifurcation (-283 ± 221 ms, average $\pm$ s.d., Figure 1D, top) and peaked around the bifurcation ($+18 \pm 145$ ms, average $\pm$ s.d., Figure 1D, bottom). Notably, half of the turn cells (51%, 211/411) became directionally selective prior to the time at which the animals' behavior diverged between left and right turns (Figure 1A, bottom) in line with premotor activity previously reported in the dSC^{10,11,13,19–21}. Furthermore, consistent with the lateralization of the dSC hemispheres for left and right turns^{10,11,19,20}, about two thirds of turn cells showed selective firing for contraversive turns relative to the recorded SC hemisphere (Figure 1E, contralateral (e.g., left turn cells in the right dSC): 69%, 283/411, ipsilateral (e.g., right turn cells in the right dSC): 31%, 128/411). Together, the properties of the turn cells identified here are consistent with the reported characteristics of motor command neurons responsible for turns in the dSC^{11–13}.

Turn cell activity is coupled to the stepping rhythm

The coordination between turn direction and stepping phase is well established in humans^{1,2}, but whether this also occurs in mice has not been verified. We therefore assessed the stepping phase of an animal as it performed left or right turns. We first characterized the stepping rhythm of mice on a transparent linear track using a bottom-view camera. As in other walking tetrapods^{22–24}, with each forepaw swing, the head shifted laterally such that a right forepaw swing was linked to a leftward head movement and vice versa, producing a left-right oscillation of the head during locomotion (Figure 2A and Figure S2). This allowed us to infer the stepping rhythm of mice on the Y-maze by monitoring the left-right oscillation of the head with a top-view camera, quantifying this oscillation as the cyclic change in head-body angle (Figure 2B, left, see Methods). We subdivided this oscillation into the following phases of the stepping cycle (Figure 2B, right): phase $1/2\pi$, when the head, aligned straight with the body, is moving left and the right forepaw is swinging forward, phase $3/2\pi$ when the head, aligned straight with the body, is moving right and the left forepaw is swinging forward and phases 0 and π , when the animal's head is pointing to the right or left, respectively, relative to the body. During locomotion, the amplitude of the head-body angle oscillated between positive and negative ~2.5 degrees (2.64 ± 0.06 deg., average $\pm$ s.e.m., here and throughout unless stated otherwise) at ~5.5 Hz (5.45 ± 0.02 Hz, ~11 steps per second). Crucially, animals reached the center point of the maze bifurcation at opposite phases of the stepping cycle depending on the direction of the turn (Figure 2C, left turns near phase 0 (2π), 6.2 ± 0.1 rad., right turns near phase π , 2.9 ± 0.1 rad., $n = 55$ sessions, $p < 0.001$ for each turn direction, Rayleigh tests, $p < 0.05$, paired Rayleigh test, mean phase difference: 3.0 rad.). Thus, at the maze bifurcation, left and right turns occur at opposite phases of the stepping cycle, consistent with other organisms^{1,2}.

Given that turns occur in coordination with the stepping rhythm, is turn cell activity in the dSC also coordinated with the stepping rhythm? We discovered that approximately half of the turn cells (47%, 195/411) exhibited firing that was significantly phase-locked to the stepping cycle (Figure 3A, example cells). This phase-locked firing was evident throughout the maze, both during straight locomotion as well as during the turn, that is, also outside of periods when turn cells showed direction-selective firing. Strikingly, left and right turn cells fired at opposite phases of the stepping cycle (Figure 3B). By focusing on turn cells with the strongest rhythmic activity (i.e., whose rhythmic activity was most strongly phase-locked to the stepping cycle: $\ln(\text{Rayleigh } Z) > \text{mean} + \text{s.d. of population}$, 25% of significantly phase-locked cells, 49/195, Figure S3A, see Methods), we found that left turn cells fired preferentially near phase $1/2\pi$, that is, when the right forepaw is

swinging forward and the head is moving left relative to the body (1.7 ± 0.2 rad., $n = 31$ cells; Figure 3C, $p < 0.05$, Rayleigh test). In contrast, right turn cells fired near the opposite phase, $3/2\pi$, that is, when the left forepaw is swinging forward and the head is moving right relative to the body (4.2 ± 0.3 rad., $n = 18$ cells; Figure 3C, $p < 0.05$, Rayleigh test). Preferred firing phases were the same regardless of the direction of the upcoming turn (Figure S3B, $r_c = 0.902$, $p < 0.001$, permutation test).

Importantly, these cells maintained the same phase preference also during periods of straight locomotion, before and after the animal engaged in the turn (Figure S3C, $r_c = 0.998$, $p < 0.001$, permutation test). The preferred firing phases of left and right turn cells were significantly different both across the population (Figure 3C, all trial periods: $p < 0.05$, permutation test, mean phase difference: 2.51 rad.; Figure 3D, straight locomotion periods: $p < 0.05$, permutation test, mean phase difference: 2.44 rad.) and in the phase differences calculated across simultaneously recorded cell pairs (Figure S3D). Thus, during locomotion, turn cell activity is rhythmically coupled to the ongoing stepping rhythm, such that left and right turn cells fire at opposite phases of the stepping cycle. Accordingly, the turn direction selectivity of a rhythmically active SC cell is revealed not only around the turn but also during straight locomotion, reflected in its preferred firing phase relative to the stepping cycle.

Turn cell activity is modulated in coordination with hippocampal sweeps

Is turn cell activity also modulated in coordination with internal representations of future paths? During navigation, the hippocampus generates internal representations of the animal's position through the spatially tuned firing of place cells^{25,26}. While place cells typically signal the animal's current location, they can also express 'non-local' activity patterns that sweep ahead of the animal toward possible left or right future paths¹⁶⁻¹⁸. To determine whether dSC turn cells fire differently when hippocampal sweeps are directed toward or away from their preferred turn direction, we recorded simultaneously from the dSC and the hippocampus (Figure S4, 167 dSC turn cells, 2642 hippocampal CA1 cells, $n = 5$ animals) as animals performed the Y-maze task. We used an established algorithm to decode position from hippocampal CA1 activity²⁷. We detected moments when, as the animal approached the bifurcation, the decoded position swept ahead, representing future paths along the left or right arm (Figure 4A and 4B, see Methods).

As expected^{28,29}, most hippocampal sweeps were directed toward the path the animal would ultimately take ('ipsi sweeps', $73.7 \pm 1.8\%$) and the rest toward the alternative path ('contra sweeps'). Contra sweeps occurred earlier relative to the turn and swept farther ahead of the animal compared to ipsi sweeps (Figure S5A and S5B). Moreover, while the onsets of both ipsi and contra sweeps occurred preferentially during late phases of the ~ 8.5 Hz (8.49 ± 0.07 Hz) hippocampal theta rhythm, contra sweeps exhibited weaker phase locking compared to ipsi sweeps (Figure S5C). Lastly, consistent with recent results in rats²⁹, the expression of contra sweeps was transiently increased following trial block switches compared to ipsi sweeps (Figure S5D).

To determine whether, for a given turn direction taken by the animal, dSC turn cell activity varied depending on the direction of hippocampal sweeps, we focused on trials with only ipsi or contra sweeps, excluding trials with sweeps in both directions and trials without sweeps. We computed the Sweep Direction Selectivity Index (SDSI, see Methods) that quantified, for turns in a given direction, the average difference in firing rate seen on trials with sweeps in the direction taken by the animal versus the opposite direction, i.e. on trials with ipsi versus trials with contra sweeps. Strikingly, the activity of a third of turn cells was significantly modulated around the time of the bifurcation depending on the direction of hippocampal sweeps (34%, 56/167, $p < 0.05$, permutation test, Figure 4C, cell 5, SDSI: $p < 0.01$, Figure 4D, cell 6, SDSI: $p < 0.05$, permutation

tests, Figure S6, additional examples). Moreover, there was a strong positive correlation between TDSI and SDSI (Figure 4E, TDSI vs. SDSI, $n = 56$ cells, Pearson $r = 0.61$, $p < 0.001$). Indeed, most of these cells fired more on average during trials with sweeps in their preferred turn direction compared to trials with sweeps in their non-preferred turn direction ('sweep-selective turn cells', 75%, 42/56). Of these, most cells (62%, 26/42) exhibited sweep direction selective firing during trials in which the animal turned in the cell's preferred direction.

To determine whether the activity of dSC neurons can be bidirectionally modulated, that is whether dSC neurons increase or decrease their firing in coordination with sweep direction, we analyzed the firing rates of each sweep-selective turn cell around the time of the bifurcation across three distinct trial types: trials with sweeps 1) only in the cell's preferred direction, 2) only in the cell's non-preferred direction, and 3) in both directions, all relative to trials without sweeps. This analysis revealed a clear bidirectional modulation of dSC firing rates relative to the no-sweep condition: sweep-selective turn cells fired more during trials with sweeps only in their preferred direction, less during trials with sweeps only in their non-preferred direction, and showed no significant change during trials with sweeps in both directions (Figure 4F, preferred/non-preferred only: $p < 0.001$, both: $p > 0.05$, 1-sample t-tests). Approximately half of the turn cells whose activity was modulated in coordination with hippocampal sweeps exhibited activity that was significantly phase-locked to the stepping rhythm (48%, 27/56), mirroring the percentage of these cells in the overall population (47%, 195/411). Together, these results reveal that the activity of dSC turn cells is modulated in coordination with hippocampal representations of future paths.

Hippocampal sweeps predict subsequent turn cell activity

Hippocampal sweeps can occur at various time points prior to the bifurcation. How early before the bifurcation do sweeps predict the activity of dSC cells at the bifurcation? To address this question, we used generalized linear models (GLMs, see Methods). As sweeps typically occurred in-phase with the hippocampal theta rhythm (Figure S5C), we divided each trial into theta cycles prior to and at the time of the bifurcation (cycles -4 through -1 and cycle 0, respectively) and, for each of these cycles, assessed whether the presence and direction of sweeps predicted dSC firing at the bifurcation (Figure 5A).

Strikingly, sweeps occurring up to 4 cycles (~ 500 ms) before the bifurcation could significantly predict dSC activity at the bifurcation (Figure 5B, $p < 0.01$, Wald z-tests). Predictive cycles (i.e. theta cycles during which sweeps significantly predicted changes in dSC activity at the bifurcation) were observed for 50% of sweep-selective turn cells and for 7.6% of all possible cycles (Figure 5B, cells: 21/42, cycles: 32/420), both of which significantly exceeded shuffled controls (Figure S7A, shuffled data: cells, $29 \pm 0.2\%$, $p = 0.001$; cycles, $4.1 \pm 0.03\%$, $p < 0.001$, permutation tests). Most predictive cycles were driven by sweeps in a cell's preferred turn direction (72%, 23/32). Of these, most cycles predicted increases in activity (Figure 5B, top, 96%, 22/23), whereas, for predictive cycles driven by sweeps in a cell's non-preferred direction (28%, 9/32) most cycles predicted decreases in activity (Figure 5B, bottom, 67%, 6/9 cycles).

Distinct dSC turn cells become directionally selective at different times prior to the bifurcation (-283 ± 221 ms, average $\pm$ s.d., Figure 1D, top). This raises the possibility that those dSC turn cells that become directionally selective earlier may also be those whose firing, at the bifurcation, is modulated in coordination with early sweeps. This was indeed the case: we found a positive correlation between the time of predictive cycles and time at which turn cells became directionally selective (Figure 5C, top, Pearson $r = 0.71$, $p < 0.01$). In other words, sweeps occurring during earlier theta cycles predicted the activity of turn cells that became directionally selective earlier, and vice versa. Thus, hippocampal sweeps can predict turn-related activity in the dSC, and the

timing of this predictability is related to the time at which individual turn cells become directionally selective.

Given that hippocampal sweeps can predict dSC activity at the bifurcation, we next asked whether an interaction in the opposite direction also occurred. That is, can dSC activity prior to the bifurcation predict hippocampal sweeps at the bifurcation? This possibility would be consistent with a direct influence of head turning behavior on hippocampal representations³⁰. To address this question, we again turned to GLMs (see Methods). Mirroring the GLM analysis above (Figure 5), we divided each trial into theta cycles prior to and at the time of the bifurcation and, for each of these cycles, assessed whether dSC activity predicted the presence and direction of hippocampal sweeps at the bifurcation.

We identified only a single instance of a significant cycle in which dSC turn cell activity predicted hippocampal sweeps at the bifurcation (Figure S8A), $p < 0.01$, Wald z-test). Additional analyses using a less conservative significance threshold yielded only weak effects that did not robustly exceed shuffled controls (Figure S8B and S8C). Thus, while dSC activity does not reliably predict hippocampal sweeps at the bifurcation, hippocampal sweeps reliably predict dSC activity.

Hippocampal sweep direction matches deviations in behavioral trajectory

dSC activity impacts turning behavior. Given that the direction of hippocampal representations of future paths predicts differences in dSC activity, we asked whether it also predicts differences in the animal's behavior. This was indeed the case.

First, for a given turn direction, turns were tighter during ipsi sweep trials compared to contra sweep trials (Figure 6A). Trajectories during contra sweep trials deviated more from a linear path and exhibited greater cumulative angular change ($IdPhi^{28}$) compared to ipsi sweep trials (Figure 6B and 6C, see Methods). Second, we observed an effect of sweep presence and direction on trial duration. Trials without sweeps had the shortest duration (median: 1.73 s), followed by ipsi sweep trials (1.83 s), contra sweep trials (1.95 s), and trials with sweeps in both directions (2.22 s; Figure S9A). Consistent with this effect, running speed was highest on trials without sweeps (median: 34.1 cm/s), followed by ipsi sweep trials (32.1 cm/s), contra sweep trials (31.2 cm/s), and trials with sweeps in both directions (27.5 cm/s; Figure S9B). Thus, the movement trajectory of the animal around the turn and its running speed reflect the presence and direction of hippocampal sweeps.

DISCUSSION

Our study shows that turn-related activity in the SC, a highly conserved vertebrate midbrain structure, is tightly coupled to ongoing locomotor behavior and modulated in coordination with hippocampal representations of possible futures.

The turn cells identified in this study share several characteristics with previously described motor command neurons for orienting movements in the SC: 1) turn cells are located in SC's intermediate and deep layers, layers whose neurons project to brainstem turning centers^{31,32}, and whose stimulation elicits turning movements^{13,14}, 2) turn cell activity primarily encodes contraversive turning movements, consistent with dSC motor signals^{10,11,19,20}, 3) turn cells become directionally selective before the time at which the animal's behavior diverges between left and right turns, consistent with premotor activity in the dSC^{10,11,13,19-21}, and 4) turn cells fire similarly at the maze bifurcation regardless of the animal's starting position on the maze, indicating that turn cell activity reflects an egocentric, displacement-based motor response. These observations suggest that the turn cell activity reported here likely reflects motor commands for turning.

Our results reveal that turn-related activity in the dSC is tightly coupled to ongoing locomotor behavior, such that left and right turn cells preferentially fire at opposite phases of the stepping cycle. To our knowledge, this is the first report of oscillatory neuronal activity in the dSC that is coupled to the stepping rhythm. Because of the opposite phases of the stepping cycle at which left and right turn cells fired, their direction preference could also be inferred during straight locomotion. The alignment of turn cell activity to opposite phases of the stepping cycle is likely to create windows of opportunity that alternatively favor left or right turns. This alignment may underlie the precise coordination observed in vertebrates between turn direction and stepping phase^{1,2} and validated here in mice (Figure 2C).

The basis for the observed rhythmic alternation in the activity of turn cells may lie in the SC's external and internal connectivity. Externally, the dSC receives direct input from motor-associated regions whose activity has been shown to be rhythmically modulated during locomotion^{33–35}. Internally, the dSC contains inhibitory connections both within and between hemispheres^{36–38}, which may support the segregation of turn cell assemblies on opposite phases of the stepping cycle. Locomotor rhythmic activity in the dSC could, however, also originate from sensory inputs^{39–41} that may oscillate in phase with locomotion. Because not all turn cells exhibited activity that was phase-locked to the stepping cycle, the neurons described in this study may constitute a distinct cell type with specific connectivity patterns.

We show that turn-related activity in the dSC is also modulated in coordination with internal representations of possible future paths. Specifically, we identified a consistent, predictive relationship between representations of possible future paths in the hippocampus and the activity of dSC neurons associated with turning behavior. Furthermore, consistent with the role of dSC activity in biasing turns, when the hippocampus represented possible paths that were not ultimately taken, movement trajectories were nonetheless biased toward those paths, as if animals partially enacted possible futures that were considered but not selected.

The possibility that the mere idea of a movement, unless inhibited, automatically triggers the movement itself has been put forward more than a century ago⁴². This suggestion has led to the proposal that all cognitive processes go on in terms of 'faint reinstatements' of overt motor acts⁴³. Indeed, numerous studies have also shown that cognitive processes can 'leak' into overt behavior^{44–48}. We propose that such leakage reflects the engagement of motor circuits by cognitive processes^{49–52}, which in turn may enable cognitive representations to guide behavior.

Previous work examining the relationship between hippocampal sweeps and behavior has primarily focused on vicarious trial and error (VTE) wherein animals' heads deviate to the left and/or right as they approach a decision point^{16,30}. While VTE still exists following hippocampal lesions^{53,54}, whether VTE could be influenced by hippocampal activity was unclear. Our results demonstrate a clear relationship between hippocampal representations of future paths and deviations in behavioral trajectory that have been associated with VTE, pointing to a potentially direct role of mental simulations of possible futures in altering the structure of behavior.

How might such mental simulations be transformed into motor output to guide behavior? Our results provide support for an integration-like mechanism in the dSC related to representations of possible futures. Previous work has demonstrated that integration is a key function of the dSC, wherein inputs related to orienting decisions drive progressive increases in activity that, upon reaching a threshold, drives action^{55–58}. Our GLM analysis shows that hippocampal sweeps occurring up to 4 theta cycles (~500 ms) before the maze bifurcation can predict dSC activity at the time of the bifurcation (Figure 5A and 5B). This timing suggests that hippocampal sweeps may be integrated to modulate dSC turning commands, perhaps in a sequential manner aligned with

the time at which individual turn cells become directionally selective (Figure 5C), to bias turns toward behaviorally relevant locations in nearby space^{59–61}. In this situation, hippocampal input to the dSC could be critical for selecting among possible turn directions.

By contrast, evidence for an interaction in the opposite direction, in which dSC activity influences hippocampal sweeps, was limited (Figure S8). This asymmetry suggests that, rather than forming a bidirectional loop, hippocampal non-local activity may preferentially influence dSC activity during online behavior. Interestingly, a recent study in mice showed that SC activity is causally related to changes in virtual heading during REM sleep¹⁰, suggesting that, conversely, SC activity may preferentially influence internal representations during offline behavior. These results raise the possibility that the direction of information flow between cognitive and motor systems may shift across online and offline states^{62,63}; during online states, when behavioral control is most critical, cognitive representations may selectively bias motor activity to guide action, whereas during offline states, motor activity may play a greater role in the formation and refinement of internal models^{10,64}.

In summary, turn-related activity in the SC, perhaps one of the best-understood motor outputs of this structure, is highly coordinated with the ongoing locomotor behavior and cognitive activity of the organism as it navigates its world. This coordination may ensure that turns occur at optimal phases of the stepping cycle and provides a potential pathway for cognitive circuits to bias behavior along internally planned trajectories.

REFERENCES

1. Patla, A.E., Prentice, S.D., Robinson, C., and Neufeld, J. (1991). Visual control of locomotion: strategies for changing direction and for going over obstacles. *J. Exp. Psychol. Hum. Percept. Perform.* 17, 603–634.
2. Patla, A.E., Adkin, A., and Ballard, T. (1999). Online steering: coordination and control of body center of mass, head and body reorientation. *Exp. Brain Res.* 129, 629–634.
3. Walter, R.M. (2003). Kinematics of 90 degrees running turns in wild mice. *J. Exp. Biol.* 206, 1739–1749.
4. Lee, D.N., and Young, D.S. (1986). Gearing action to the environment. In *Generation and Modulation of Action Patterns* (Springer Berlin Heidelberg), pp. 217–230.
5. Tuhkanen, S., Pekkanen, J., Wilkie, R.M., and Lappi, O. (2021). Visual anticipation of the future path: Predictive gaze and steering. *J. Vis.* 21, 25.
6. Basso, M.A., and May, P.J. (2017). Circuits for action and cognition: A view from the superior colliculus. *Annu. Rev. Vis. Sci.* 3, 197–226.
7. Isa, T., Marquez-Legorreta, E., Grillner, S., and Scott, E.K. (2021). The tectum/superior colliculus as the vertebrate solution for spatial sensory integration and action. *Curr. Biol.* 31, R741–R762.
8. Felsen, G., and Mainen, Z.F. (2012). Midbrain contributions to sensorimotor decision making. *J. Neurophysiol.* 108, 135–147.
9. Felsen, G., and Mainen, Z.F. (2008). Neural substrates of sensory-guided locomotor decisions in the rat superior colliculus. *Neuron* 60, 137–148.
10. Senzai, Y., and Scanziani, M. (2024). The brain simulates actions and their consequences during REM sleep. *bioRxiv.org*. <https://doi.org/10.1101/2024.08.13.607810>.
11. Sharp, S.L., Shin, J., Martins, D.M., Jones, K., and Niell, C.M. (2025). Neural dynamics in superior colliculus of freely moving mice. *Cell Rep.* 44, 116284.
12. Wilson, J.J., Alexandre, N., Trentin, C., and Tripodi, M. (2018). Three-dimensional representation of motor space in the mouse superior colliculus. *Curr. Biol.* 28, 1744–1755.e12.
13. Masullo, L., Mariotti, L., Alexandre, N., Freire-Pritchett, P., Boulanger, J., and Tripodi, M. (2019). Genetically defined functional modules for spatial orienting in the mouse superior colliculus. *Curr. Biol.* 29, 2892–2904.e8.
14. Zahler, S.H., Taylor, D.E., Wright, B.S., Wong, J.Y., Shvareva, V.A., Park, Y.A., and Feinberg, E.H. (2023). Hindbrain modules differentially transform activity of single collicular neurons to coordinate movements. *Cell* 186, 3062–3078.e20.
15. Cooper, B.G., Miya, D.Y., and Mizumori, S.J. (1998). Superior colliculus and active navigation: role of visual and non-visual cues in controlling cellular representations of space. *Hippocampus* 8, 340–372.
16. Johnson, A., and Redish, A.D. (2007). Neural ensembles in CA3 transiently encode paths forward of the animal at a decision point. *J. Neurosci.* 27, 12176–12189.
17. Kay, K., Chung, J.E., Sosa, M., Schor, J.S., Karlsson, M.P., Larkin, M.C., Liu, D.F., and Frank, L.M. (2020). Constant sub-second cycling between representations of possible futures in the hippocampus. *Cell* 180, 552–567.e25.
18. Vollan, A.Z., Gardner, R.J., Moser, M.-B., and Moser, E.I. (2025). Left-right-alternating theta sweeps in entorhinal-hippocampal maps of space. *Nature* 639, 995–1005.
19. Wurtz, R.H., and Goldberg, M.E. (1971). Superior colliculus cell responses related to eye movements in awake monkeys. *Science* 171, 82–84.
20. Schiller, P.H., and Stryker, M. (1972). Single-unit recording and stimulation in superior colliculus of the alert rhesus monkey. *J. Neurophysiol.* 35, 915–924.
21. Glimcher, P.W., and Sparks, D.L. (1992). Movement selection in advance of action in the superior colliculus. *Nature* 355, 542–545.

22. Kafkafi, N., Levi-Havusha, S., Golani, I., and Benjamini, Y. (1996). Coordination of side-to-side head movements and walking in amphetamine-treated rats: a stereotyped motor pattern as a stable equilibrium in a dynamical system. *Biol. Cybern.* 74, 487–495.
23. Goyens, J., and Aerts, P. (2018). Head stabilisation in fast running lizards. *Zoology (Jena)* 127, 114–120.
24. Mulavara, A.P., Verstraete, M.C., and Bloomberg, J.J. (2002). Modulation of head movement control in humans during treadmill walking. *Gait Posture* 16, 271–282.
25. O'Keefe, J., and Dostrovsky, J. (1971). The hippocampus as a spatial map. Preliminary evidence from unit activity in the freely-moving rat. *Brain Res.* 34, 171–175.
26. Wilson, M.A., and McNaughton, B.L. (1993). Dynamics of the hippocampal ensemble code for space. *Science* 261, 1055–1058.
27. Denovellis, E.L., Gillespie, A.K., Coulter, M.E., Sosa, M., Chung, J.E., Eden, U.T., and Frank, L.M. (2021). Hippocampal replay of experience at real-world speeds. *Elife* 10. <https://doi.org/10.7554/eLife.64505>.
28. Papale, A.E., Zielinski, M.C., Frank, L.M., Jadhav, S.P., and Redish, A.D. (2016). Interplay between hippocampal sharp-wave-ripple events and vicarious trial and error behaviors in decision making. *Neuron* 92, 975–982.
29. Comrie, A.E., Monroe, E.J., Kahn, A.E., Denovellis, E.L., Joshi, A., Guidera, J.A., Krausz, T.A., Berke, J.D., Daw, N.D., and Frank, L.M. (2024). Hippocampal representations of alternative possibilities are flexibly generated to meet cognitive demands. *bioRxiv*org. <https://doi.org/10.1101/2024.09.23.613567>.
30. Redish, A.D. (2016). Vicarious trial and error. *Nat. Rev. Neurosci.* 17, 147–159.
31. Cregg, J.M., Leiras, R., Montalant, A., Wanken, P., Wickersham, I.R., and Kiehn, O. (2020). Brainstem neurons that command mammalian locomotor asymmetries. *Nat. Neurosci.* 23, 730–740.
32. Usseglio, G., Gatier, E., Heuzé, A., Hérent, C., and Bouvier, J. (2020). Control of orienting movements and locomotion by projection-defined subsets of brainstem V2a neurons. *Curr. Biol.* 30, 4665–4681.e6.
33. Drew, T. (1993). Motor cortical activity during voluntary gait modifications in the cat. I. Cells related to the forelimbs. *J. Neurophysiol.* 70, 179–199.
34. Beloozerova, I.N., Nilaweera, W.U., Viana Di Prisco, G., and Marlinski, V. (2023). Signals from posterior parietal area 5 to motor cortex during locomotion. *Cereb. Cortex* 33, 1014–1043.
35. Hirokane, K., Nakamura, T., Terashita, T., Kubota, Y., Hu, D., Yagi, T., Graybiel, A.M., and Kitsukawa, T. (2024). Representation of rhythmic chunking in striatum of mice executing complex continuous movement sequences. *Cell Rep.* 43, 114312.
36. Sprague, J.M. (1966). Interaction of cortex and superior colliculus in mediation of visually guided behavior in the cat. *Science* 153, 1544–1547.
37. Munoz, D.P., and Istvan, P.J. (1998). Lateral inhibitory interactions in the intermediate layers of the monkey superior colliculus. *J. Neurophysiol.* 79, 1193–1209.
38. Meredith, M.A., and Ramoa, A.S. (1998). Intrinsic circuitry of the superior colliculus: pharmacophysiological identification of horizontally oriented inhibitory interneurons. *J. Neurophysiol.* 79, 1597–1602.
39. Meredith, M.A., and Stein, B.E. (1986). Visual, auditory, and somatosensory convergence on cells in superior colliculus results in multisensory integration. *J. Neurophysiol.* 56, 640–662.
40. Maeda, M., Shibazaki, T., and Yoshida, K. (1979). Labyrinthine and visual inputs to the superior colliculus neurons. *Prog. Brain Res.* 50, 735–743.
41. Chinta, S., and Pluta, S.R. (2025). Whisking and locomotion are jointly represented in superior colliculus neurons. *PLoS Biol.* 23, e3003087.
42. James, W. (1890). *The Principles of Psychology* (Vol. 2) (Henry Holt & Company).
43. Watson, J.B. (1913). Psychology as the behaviorist views it. *Psychol. Rev.* 20, 158–177.

44. Euston, D.R., and McNaughton, B.L. (2006). Apparent encoding of sequential context in rat medial prefrontal cortex is accounted for by behavioral variability. *J. Neurosci.* 26, 13143–13155.
45. Cowen, S.L., and McNaughton, B.L. (2007). Selective delay activity in the medial prefrontal cortex of the rat: contribution of sensorimotor information and contingency. *J. Neurophysiol.* 98, 303–316.
46. Korbisch, C.C., Apuan, D.R., Shadmehr, R., and Ahmed, A.A. (2022). Saccade vigor reflects the rise of decision variables during deliberation. *Curr. Biol.* 32, 5374–5381.e4.
47. Calalo, J.A., Ngo, T.T., Sullivan, S.R., Strand, K., Buggeln, J.H., Lokesh, R., Roth, A.M., Carter, M.J., Kurtzer, I.L., and Cashaback, J.G.A. (2025). Online movements reflect ongoing deliberation. *J. Neurosci.* 45, e1913242025.
48. Cazettes, F., Reato, D., Augusto, E., Steinfeld, R., Renart, A., and Mainen, Z.F. (2025). Facial expressions in mice reveal latent cognitive variables and their neural correlates. *Nat. Neurosci.* 28, 2310–2318.
49. Rizzolatti, G., Riggio, L., Dascola, I., and Umiltà, C. (1987). Reorienting attention across the horizontal and vertical meridians: evidence in favor of a premotor theory of attention. *Neuropsychologia* 25, 31–40.
50. Craighero, L., and Rizzolatti, G. (2005). The premotor theory of attention. In *Neurobiology of Attention* (Elsevier), pp. 181–186.
51. Jeannerod, M. (2001). Neural simulation of action: a unifying mechanism for motor cognition. *Neuroimage* 14, S103–9.
52. Tse, P.U. (2025). The disencapsulated mind: A premotor theory of human imagination. *Psychol. Rev.* 132, 895–915.
53. Clark, B.J., Hines, D.J., Hamilton, D.A., and Whishaw, I.Q. (2005). Movements of exploration intact in rats with hippocampal lesions. *Behav. Brain Res.* 163, 91–99.
54. Papale, A.E. (2015). Hippocampal Representations on the Spatial Delay Discounting Task.
55. Dorris, M.C., Paré, M., and Munoz, D.P. (1997). Neuronal activity in monkey superior colliculus related to the initiation of saccadic eye movements. *J. Neurosci.* 17, 8566–8579.
56. Sparks, D., Rohrer, W.H., and Zhang, Y. (2000). The role of the superior colliculus in saccade initiation: a study of express saccades and the gap effect. *Vision Res.* 40, 2763–2777.
57. Paré, M., and Hanes, D.P. (2003). Controlled movement processing: superior colliculus activity associated with countermanded saccades. *J. Neurosci.* 23, 6480–6489.
58. Jantz, J.J., Watanabe, M., Everling, S., and Munoz, D.P. (2013). Threshold mechanism for saccade initiation in frontal eye field and superior colliculus. *J. Neurophysiol.* 109, 2767–2780.
59. Tang, W., Mei, X., Harvey, R.E., Carbajal-Leon, E., Netzer, T., Chang, H., Oliva, A., and Fernandez-Ruiz, A. (2025). Goal-directed hippocampal theta sweeps during memory-guided navigation. *bioRxiv.org*. <https://doi.org/10.1101/2025.08.26.672489>.
60. Yu, C., Ji, Z., Ormond, J., O’Keefe, J., and Burgess, N. (2025). Hippocampal theta sweeps indicate goal direction. *bioRxiv*. <https://doi.org/10.1101/2025.08.21.671551>.
61. Vollan, A.Z., Schellenberger, M.F., Moser, M.-B., and Moser, E.I. (2026). Attention-like regulation of theta sweeps in the brain’s spatial navigation circuit. *bioRxiv*. <https://doi.org/10.64898/2026.01.27.702083>.
62. Buzsáki, G. (1989). Two-stage model of memory trace formation: a role for “noisy” brain states. *Neuroscience* 31, 551–570.
63. Hasselmo, M.E. (1999). Neuromodulation: acetylcholine and memory consolidation. *Trends Cogn. Sci.* 3, 351–359.
64. Hobson, J.A., Hong, C.C.-H., and Friston, K.J. (2014). Virtual reality and consciousness inference in dreaming. *Front. Psychol.* 5, 1133.
65. Pachitariu, M., Sridhar, S., and Stringer, C. (2023). Solving the spike sorting problem with Kilosort. *bioRxiv*. <https://doi.org/10.1101/2023.01.07.523036>.

66. Mathis, A., Mamidanna, P., Cury, K.M., Abe, T., Murthy, V.N., Mathis, M.W., and Bethge, M. (2018). DeepLabCut: markerless pose estimation of user-defined body parts with deep learning. *Nat. Neurosci.* 21, 1281–1289.
67. LorenFrankLab (2024). track_linearization: 2D to 1D position linearization using HMMs. https://github.com/LorenFrankLab/track_linearization.
68. Siapas, A.G., Lubenov, E.V., and Wilson, M.A. (2005). Prefrontal phase locking to hippocampal theta oscillations. *Neuron* 46, 141–151.
69. Franklin, K.B.J., and Paxinos, G. (2007). *The mouse brain in stereotaxic coordinates* 3rd ed. (Academic Press).

METHODS

Experimental model and animals

Neural activity was recorded from twelve male C57BL/6J mice (Jackson Laboratory #000664, 3.5-8 months old). All mice were housed on a reversed cycle (light/dark cycle 12/12h). Mice were housed with littermates before experimentation and singly housed in enriched cages during food-restriction and experimental protocols. All experimental procedures were conducted in accordance with the regulations of the University of California San Francisco Institutional Animal Care and Use Committee (IACUC).

Behavioral training and electrode implantation

The implantation procedure was performed in two phases on two distinct days. First, the implantation of the headplate, and second, the implantation of the silicon probe(s). Mice were implanted under 2% isoflurane anesthesia. The headplate was placed on the skull with dental cement (Unifast LC, GC America; Optibond Universal, Kerr Dental). After post-operative recovery of five days in their home cage, mice were deprived of food to 85% of their baseline weight and pretrained to run on an elevated 1 m linear track for liquid reward (sweetened evaporated soymilk). This training was done to familiarize the mice with running on a track and accessing reward ports. After the mice alternated between the reward ports reliably on the linear track (1-3 days), they were pretrained on the Y-maze task (3-5 days, see *Y-maze task and trial segmentation*). After successful pretraining on the Y-maze, mice underwent the silicon probe implantation surgery. Silicon probes were mounted on a 3D-printed moveable microdrive to record unilaterally or bilaterally from the superior colliculus (SC). In five out of the twelve mice, a silicon probe was also implanted in the dorsal CA1 region of the hippocampus in the opposite hemisphere. The types of probes used included: Single-shank probes (Diagnostic Biochips (DBC) P64-4, 64 channels per shank, 20 μ m interchannel pitch), four-shank probes (DBC P64-1, 250 μ m inter-shank distance, 16 channels per shank, 20 μ m interchannel pitch, DBC P128-6, 150 μ m inter-shank distance, 32 channels per shank, 25 μ m interchannel pitch), and eight-shank probes (DBC P128-8, 150 μ m inter-shank distance, 16 channels per shank, 30 μ m interchannel pitch). Probe shanks were inserted through a small craniotomy at the coordinates described in the *Electrode insertion, electrode coordinates, histology and recording site assessment* section below. The shanks were initially lowered to 0.7-1.2 mm (SC) and 0.6 mm (CA1) below brain surface and the microdrive was then fixed, via dental cement, to the headplate. A ground electrode (0.005" diameter stainless wire, A-M systems) was inserted in the cerebellum. After post-operative recovery, mice were placed back on the Y-maze training protocol for 1-5 days while the shanks were lowered to the intermediate and deep SC (dSC) and CA1. This protocol ensured that peak Y-maze performance coincided with the time at which the recording sites reached their targets. Mice remained on food restriction for the remainder of the experiment.

Electrophysiological recordings and video tracking

The data presented in this paper are from three to eight 45-120 min sessions per mouse on the Y-maze task. For the stepping analysis, two implanted mice also ran on a transparent linear track (Figure 2, Figure S2, see *Transparent linear track task and stepping analysis*). Electrophysiological data were acquired using an Intan RHD2000 system (Intan Technologies LLC) band-pass filtered between 0.1 Hz and 7.5 kHz and digitized at 20 kHz. Spike sorting was

performed semi-automatically, using Kilosort 2.0 and Phy⁶⁵. For the Y-maze task, video data was acquired at 60 frames per second using a CMOS camera (Basler acA1300-200um) placed above the maze. For the transparent linear track task, video was acquired at 124 frames per second using a CMOS camera (Basler acA1300-200um) with a wide-angle lens placed below the track. A machine-learning algorithm, DeepLabCut⁶⁶, was trained to track features of the probe housing and distinct body parts of the mice (Y-maze: microdrive probe base (head), left and right ears, base of the tail, transparent linear track: forepaws, under-head/chin area). For position analysis and trial segmentation on the Y-maze, the probe base position (head position) was used as the actual position of the mouse.

Y-maze task and trial segmentation

Mice navigated an elevated Y-maze with equally spaced arms (arm length: 45 cm, arm width: 5 cm) in search of liquid food reward (sweetened evaporated soymilk), which was dispensed from two out of three ports at the ends of each arm. The task consisted of six to eight 50-80 trial blocks, with the locations of the active ports switching in an uncued manner between blocks. The active ports were chosen randomly within each block, with no block containing the same two active ports as the previous block. Animals could not visit the same port on two consecutive trials in order to receive a reward, so mice learned to alternate between the active ports. For trial segmentation, we first used the *track linearization* package⁶⁷ to map the animal's position on the Y-maze to a simplified one-dimensional representation with a corresponding maze arm identifier. We then defined a track "position threshold" at 70% of the length of each maze arm (31.5 cm measured outward from the center of the maze or 13.5 cm measured inward from the end of the arm). Trials were defined as periods from when the animal, after leaving a reward port, crossed the position threshold of one arm, made a turn, and crossed the position threshold of another arm without entering any other arms. The time at which an animal reached the bifurcation was defined as the time at which the animal's linearized head position transitioned from one arm to another at the center point of the maze. Animals took ~1.5 seconds to complete each trial (median: 1.57 s, IQR: 1.27-1.98 s, n = 24207 trials across 12 mice).

Head angle relative to the track and time of behavioral divergence

Head angle relative to the track was defined as the angle between the animal's head direction vector (given by the vector perpendicular to the line connecting the ears) and the midline of the track segment from which the animal approached the bifurcation. The time of behavioral divergence was defined as the time at which the animal's head angle relative to the track significantly diverged between left and right turns. To determine the average time of behavioral divergence across sessions and animals (n = 55 sessions, 12 animals, Figure 1A, bottom), we calculated a Behavioral Divergence Index (BDI) as the difference between session-averaged head angles relative to the track for left and right turn trials. BDI was calculated within 200 ms windows advanced in 20 ms steps from 0.85 s before to 0.85 s after the time at which the animal reached the bifurcation. BDI significance was assessed using a permutation test: for each window, a null distribution was generated by randomly permuting left and right turn labels across sessions (10,000 permutations) and recalculating BDI from the resulting averages. Windows were considered significant only when the observed BDI was more extreme than all shuffled values (p

< 1/10,000). The time of behavioral divergence was defined as the earliest time window belonging to a continuous sequence of such significant windows.

Head-body oscillation and phase estimation

Head-body angle was defined as the angle between the animal's head direction vector (given by the vector perpendicular to the line connecting the ears) and its body vector (given by the line connecting the midpoint between the ears and the base of the tail). To determine the average frequency of the locomotor head-body oscillation, head-body angle traces for each trial were extracted, smoothed with a Savitzky–Golay filter (50 ms, first-order) and bandpass filtered between 3 and 12 Hz (Butterworth, third-order). For all other analyses, head-body angle traces were smoothed and bandpass filtered between 4 and 8 Hz (Butterworth, third-order). Furthermore, trials were included only if they exhibited sufficient power in the head-body angle oscillation signal. To quantify this, for each trial in a given session, power spectral density was estimated using Welch's method and the peak spectral power within the 4-8 Hz range was identified. To ensure that trials had sufficient power for spike-phase estimation, trials whose peak power fell more than one standard deviation below the mean peak power across trials in that session were excluded from analysis.

To estimate the phase of the head-body oscillation at the time of discrete behavioral or neural events (i.e., turns or spikes), we used a Hilbert transform-based interpolation approach. Head-body angle traces were first smoothed and filtered as described above. Peaks and troughs of the filtered signal were then identified (*scipy.signal.find_peaks*, prominence = 1° amplitude) and the instantaneous phase was obtained using the Hilbert transform. The oscillatory signal was then segmented into half-cycles (peak-to-trough and trough-to-peak). Event times were assigned a phase by identifying the half-cycle in which the event occurred and linearly interpolating the Hilbert phase within that interval. Events occurring exactly at peaks or troughs were assigned phases of 0 and π , respectively. Full oscillatory cycles, defined by successive peaks, with durations outside the expected range (i.e., cycles faster than 8 Hz or slower than 4 Hz) were excluded from analysis.

Transparent linear track task and stepping analysis

For stepping analysis, implanted mice ($n = 2$) shuttled back and forth on a transparent linear track (width: 12.7 cm (x), length: 91.4 cm (y)) for liquid food reward (sweetened evaporated soymilk). Video was recorded from below using a wide-angle lens and DeepLabCut was used to track the animals' forepaws and under-head/chin area (head position) for analyzing locomotor dynamics. Due to video distortions at the edges of the track associated with the wide-angle lens, only periods when the mouse was directly above the camera (within a 12.7 cm (x) by 32 cm (y) area) and locomoting uninterrupted were analyzed.

Forepaw swings were defined as periods when either forepaw advanced in the y-direction at speeds greater than 37.65 cm/s (3 pixels/frame). These times were then used as temporal references to examine lateral movements of the head during forepaw swings. Head position in the x-direction was filtered between 4 and 10 Hz (Butterworth, third-order). During locomotion, the head shifted laterally during each forepaw swing, such that a right forepaw swing was linked to a leftward head movement and vice-versa (Figure 2A). Notably, this same pattern (i.e.,

contralateral coupling between forepaw swings and head movements) has been well characterized in walking rats during exaggerated lateral head movements²², as well as in other walking tetrapods^{23,24}. To quantify the direction of head movement during forepaw swings, we aligned the filtered head position in the x-direction to the middle of each forepaw swing and calculated the instantaneous change in head position ($d(\text{head } x \text{ pos. filt.})/dt$, Figure S2B). In both mice, left forepaw swings were associated with positive (i.e., rightward) changes in head position while right forepaw swings were associated with negative (i.e., leftward) changes in head position (Figure S2B and S2C). Furthermore, when both the forepaw and head movement signals were filtered between 4 and 10 Hz, their instantaneous frequencies were correlated across mice (Figure S2D). Thus, the left-right oscillation of the head characteristic of locomotion provides a reliable proxy for the stepping cycle in mice.

Electrode insertion, electrode coordinates, histology and recording site assessment

In six mice, the right SC was targeted; in four mice, the left SC was targeted; and in two mice both the right and the left SC were targeted simultaneously. Lambda was used as the skull reference for all SC craniotomies, while bregma was used for hippocampal craniotomies. In five mice, the SC was targeted by drilling a craniotomy at anteroposterior (AP, from lambda): 0 mm, mediolateral (ML): ± 2.0 mm. The silicon probe(s) were then inserted at a 30° angle from vertical (i.e., a 30° medial tilt along the coronal plane). In six mice, the SC was targeted by a craniotomy at AP: 0 mm, ML: ± 2.5 mm and the silicon probe(s) were inserted at a 32° angle from vertical. In one mouse, the SC was targeted by a craniotomy at AP: 0 mm, ML: -1.0 mm and the silicon probe was inserted vertically with a 10° anterior tilt. For all SC implantations, probes were inserted along the anterior-posterior axis of the SC. The CA1 region of the hippocampus was targeted in the opposite hemisphere by a craniotomy at AP: -2.0 mm from bregma, ML: ± 1.5 mm, and the silicon probe was inserted vertically at a 45° angle relative to the sagittal plane along the long axis of the hippocampus. Prior to implantation, the backs of all probe shanks were coated with Dil solution (Fisher, V22885) for probe track localization using a fluorescence microscope. The shanks were initially lowered to 0.7-1.2 mm (SC) and 0.6 mm (CA1) below the surface of the brain (dorsoventral, DV) and advanced over consecutive days to reach the dSC and cell body layer of CA1. The dSC was targeted using DV coordinates and CA1 was targeted using DV coordinates and physiological signatures such as theta and sharp-wave ripples.

Once the target regions were reached, dSC probes were advanced between sessions (days) to record from different neurons while CA1 probes were slightly advanced between sessions to maintain maximal units for decoding. For the dSC implantations in which the probe was inserted at an angle ($n = 13$ implantations from 11 mice (9 unilateral, 2 bilateral)), the average depth for the first dSC recording was DV: -1.20 ± 0.11 mm measured from the topmost channel of the probe while the average depth for the last dSC recording was DV: -2.83 ± 0.07 mm measured from the bottommost channel of the probe (Figure S1B, left). For the single vertical implantation, the recordings spanned from DV: -0.88 to -2.30 mm (Figure S1B, right). For CA1, average recording depths were DV: -1.00 ± 0.11 mm and DV: -1.39 ± 0.10 , respectively (Figure S1B, right).

For anatomical analysis, mice were perfused transcardially with phosphate-buffered saline (PBS) and then with 4% paraformaldehyde (PFA) in PBS. Brains were extracted, stored in 4% PFA overnight at 4°C and subsequently cut with a vibratome to 100 µm thick coronal sections. Slices were mounted in a Vectashield mounting medium containing DAPI (Vector Laboratories H1500) and fluorescence images were acquired with an Olympus MVX10 MacroView microscope. Probe tracks were identified using the Dil signal.

Turn Direction Selectivity Index (TDSI) and turn cells

Turn selectivity was quantified using a Turn Direction Selectivity Index (TDSI), calculated as the difference between the average firing rates during right and left turns divided by their sum ((Right FR - Left FR) / (Right FR + Left FR)) over a 200 ms window centered on the time of the cell's peak firing rate. Thus, right turn cells were defined by positive TDSI values and left turn cells were defined by negative TDSI values. To obtain firing rates, spike trains were converted to binary vectors and smoothed using convolution with a Gaussian kernel (50 ms window width). The resulting spike density functions were normalized by the kernel width to express firing rate in spikes per second and averaged across trials. To determine the time window for the TDSI calculation, we first defined a wide window of ± 0.5 s around the time at which the animal reached the bifurcation and, for each neuron, identified within this window the time of peak firing rate for either left or right turns (whichever was greater), using the neuron's average firing rate across all arms. TDSI was then computed across each maze arm within a narrow window of ± 0.1 s (200 ms) centered on this peak firing time.

TDSI significance was assessed using a permutation test. For each neuron and maze arm, left and right turn trials were pooled together and randomly reassigned into two groups whose sizes matched the original sample sizes. TDSI was then recomputed within the narrow window centered on the same actual peak firing time. This procedure was repeated 1000 times to generate a shuffled distribution of TDSI values. For right turn cells, the p-value was defined as the fraction of shuffled TDSI values that were greater than or equal to the observed TDSI, or less than or equal to the observed TDSI for left turn cells. Turn cells were defined as neurons whose TDSI retained the same sign and remained significant ($p < 0.05$) regardless of which maze arm the animal approached the bifurcation from.

Instantaneous TDSI and significant TDSI onset

We computed instantaneous TDSI across the entire trial period⁹ to determine the time at which turn cells became directionally selective (significant TDSI onset). TDSI was calculated within 200 ms windows advanced in 20 ms steps from 0.85 s before to 0.85 s after the time at which the animal reached the bifurcation. TDSI significance was assessed with a permutation test: for each window, left and right turn labels were randomly permuted across trials and TDSI was recomputed. This process was repeated 500 times to generate a null distribution for each window. Windows were considered significant when the observed TDSI was more extreme than all shuffled values ($p < 1/500$). Significant TDSI onset was defined as the earliest time point at which a turn cell exhibited two consecutive significant TDSI windows in the expected direction (e.g., for a left turn cell, two consecutive significant TDSI windows with negative TDSI values).

Phase-locking analysis of turn cell activity to the head-body oscillation cycle

The phase of the head-body oscillation cycle at the time of each spike was estimated using a Hilbert transform-based interpolation approach (see *Head-body oscillation and phase estimation*). A cell's preferred firing phase was defined as the circular mean of all spikes pooled across all trial periods (see *Circular statistics*). Preferred firing phases were also calculated separately for spikes occurring during left or right turns, and straight locomotor periods (Figure S3B and S3C). To obtain firing rates relative to the head-body oscillation cycle, spike counts were first divided by the total number of cycles and then by the average time per bin based on the average head-body oscillation frequency across animals (5.5 Hz). Phase locking of turn cell activity was assessed via a Rayleigh's test for circular uniformity (see *Circular statistics*).

Head-body cycles from turn

The unwrapped phase of the head-body oscillation signal was obtained for each trial via the Hilbert transform and referenced to the cycle containing the turn. Specifically, the nearest peak of the head-body oscillation signal relative to the turn was used as cycle zero by subtracting its phase value from the unwrapped signal. Preceding or following peaks were defined as cycles -1, +1, etc. Spike times were then assigned a phase value by linearly interpolating the unwrapped phase at each spike time. Phase-aligned spike counts were then binned (bin width = $\pi/3$) and divided by the number of trials. To obtain firing rates, these spike counts per trial were further divided by the average time per bin based on the average head-body oscillation frequency across animals (5.5 Hz). Analyses were done separately for left and right turn trials.

Straight locomotor periods

Straight locomotor periods were defined as periods in which the animal traversed the outer region of the maze, away from the bifurcation. Specifically, we defined a region comprising the outer 50% of the previously defined 70% arm segment length used for trial segmentation (i.e. from 13.5 to 29.25 cm inward from the end of the arm; see *Y-maze task and trial segmentation*). Preferred firing phases during straight locomotion were estimated using only spikes that occurred when the animal was in these regions of the maze, during both inward and outward traversals.

Circular statistics

To quantify the relationship between rhythmic turn cell firing and the head-body oscillation cycle, we employed several circular statistical analyses. Spike times were first converted to phase (see *Head-body oscillation and phase estimation*). The preferred firing phase was calculated as the circular mean of all spike phases, determined by the angle of the mean resultant vector. To determine the strength of phase-locking and its statistical significance, we calculated a Rayleigh's Z statistic for each cell, a standard method for assessing phase-locking between neuronal spiking and local field potential oscillations⁶⁸. Rayleigh's Z statistic uses the formula $Z = nR^2$, where n is the number of spikes and R is the mean resultant length. The associated p-value of the Rayleigh's test was estimated as $p = e^{-Z}$; this approximation quantifies the probability that the observed clustering of spikes during the stepping cycle occurred by chance, assuming a null hypothesis of uniform circular distribution. For $n > 50$, $p = e^{-Z}$ is adequate⁶⁸. To account for multiple comparisons across the population of 411 turn cells, a Bonferroni-corrected significance threshold

was applied ($\alpha = 0.05/411$). A cell was considered significantly phase-locked if its p-value was less than this adjusted alpha level (47% of turn cells, 195/411).

To identify the population of turn cells with the strongest rhythmic activity (Figure 3C), we employed a two-step filtering process. First, we identified all cells with a Rayleigh Z statistic meeting a nominal significance threshold of $p < 0.05$ (69% of turn cells, 282/411). To normalize the distribution of these values, we calculated the natural log of their Rayleigh Z scores ($\ln(Z)$). We then defined a 'high-strength' subpopulation consisting of cells whose $\ln(Z)$ exceeded the mean + SD of this distribution (mean: 3.11, SD: 1.45, mean + SD: 4.56). This secondary filter ensured that subsequent phase analyses focused on neurons with consistently high phase-locking strength (Figure 3A).

To assess whether the preferred firing phases of left and right turn cells within this high-strength population were significantly different (Figure 3C), we calculated the mean circular difference between the two groups. The observed difference was computed as the angular distance between the circular means of each group, mapped to the range $[0, \pi]$ to represent the shortest distance on the circle. Statistical significance was assessed using a permutation test ($n = 1,000$ shuffles). For each shuffle, the turn cell identities (left or right) were randomly shuffled while maintaining the original sample sizes and a null distribution of angular differences was generated. The p-value was defined as the proportion of shuffled iterations where the angular difference was greater than or equal to the observed difference. Differences were considered significant if $p < 0.05$.

To assess differences between paired phase measurements (e.g., stepping phase at the time at which the animal reached the bifurcation for left vs. right turns; Figure 2C), we computed circular differences between paired observations and applied a Rayleigh test ($p = e^{-Z}$, as above) to assess whether these differences were non-uniformly distributed. Paired Rayleigh tests were considered significant if $p < 0.05$, indicating a consistent directional shift between conditions.

Correlations between distributions of preferred firing phases (Figure 3B, left vs. right turn trials, Figure 3C, all trial periods vs. straight locomotor periods) were quantified using a circular correlation coefficient (r_c) that is robust to phase wrapping. This metric measures the degree to which deviations from the mean phase in one distribution co-vary with deviations from the mean phase in the other. Correlations were computed using the *circcorcoeff* function in *astropy.stats* in Python. Significance was assessed using a permutation test ($n = 10,000$ shuffles). For each shuffle, one of the input variables was randomly shuffled and a null distribution of resulting r_c values was generated. The p-value was defined as the proportion of shuffled iterations where the r_c was greater than or equal to the observed r_c .

Spatial decoding from hippocampal activity

We used an established state space algorithm from Denovellis et al., 2021²⁷ to decode spatial position from populations of clustered hippocampal neurons. Briefly, for each session we built an encoding model to relate hippocampal spiking to the animal's linearized head position in 1 ms bins. We then decoded position from hippocampal spiking in 0.5 cm linear position bins and 1 ms temporal bins. For cross-validation, we trained the encoding model on 75% of the data in each

session and tested it on the remaining 25%, repeating this across four folds. We used a single-state continuous, causal decoding model²⁷ in which the continuous state was modeled by a random walk transition matrix with a 9.6 cm movement variance. To obtain the decoded position, the posterior probability distribution of position was first smoothed across spatial bins (i.e., along the position axis) using a 2.5 cm moving window. The decoded position in each time bin was then defined as the location of the peak of this smoothed posterior. For each time bin, we also classified the animal's actual and decoded positions into one of three maze arms.

Identification of hippocampal sweeps from decoded position

Sweep times were identified as times during trial periods (i.e., during locomotion) in which the decoded position was in a maze arm distinct to the animal's actual position for at least 15 consecutive time bins (15 ms). This duration requirement was used to prevent rapid, spurious jumps in the decoded position from being identified as sweeps. Only sweeps that occurred prior to the bifurcation were analyzed. Sweeps were classified as ipsi sweeps if they proceeded into the maze arm that the animal would eventually take or contra sweeps if they proceeded into the alternative arm. For trial-based analyses, trials were classified as ipsi sweep trials (containing only ipsi sweeps (1 or more)), contra sweep trials (containing only contra sweeps), trials with sweeps in both directions (containing both ipsi and contra sweeps), or trials without sweeps.

To compare the sweeps identified here in mice to those observed in rats, we extracted several key features of ipsi and contra sweeps including their timing relative to the hippocampal theta cycle, how far they swept ahead of the animal, and their density around the time of trial block switches (Figure S5). Mirroring hippocampal sweeps observed in rats, ipsi and contra sweeps occurred preferentially during late phases of the theta cycle (0 to π rad., Figure S5C)^{17,18,29} and swept one body length ahead of the animal on average (~9 cm here, Figure S5B, ~25 cm in rats^{18,29}). Furthermore, consistent with a recent report in rats²⁹, we observed a transient increase in the density of contra sweeps relative to ipsi sweeps after block switches (Figure S5D). Thus, the hippocampal sweeps reported here in mice exhibit strikingly similar features to those in rats.

Sweep direction selectivity index (SDSI)

To investigate the relationship between dSC turn cell firing and sweep direction, we took advantage of the fact that, for a given turn direction, some trials contain ipsi sweeps and others contain contra sweeps. Thus, for trials in a given turn direction, we could assess whether the activity of individual dSC turn cells varied depending on the direction of hippocampal sweeps. As this analysis necessarily required trials with only ipsi or contra sweeps, we excluded trials with sweeps in both directions and trials without sweeps.

For each dSC turn cell, we then computed a sweep direction selectivity index (SDSI) around the bifurcation separately for all left turn trials and all right turn trials. SDSI was calculated as the difference between the average spike count during all trials with right sweeps and the average spike count during all trials with left sweeps, divided by their sum ((Right sweep spike count - Left sweep spike count) / (Right sweep spike count + Left sweep spike count)), where spike counts were taken from a 200 ms window centered on the time that the animal crossed the bifurcation ($t = 0$). Thus, for a given turn direction, cells that fired more at the bifurcation during trials with right

sweeps were defined by positive SDSI values and cells that fired more during trials with left sweeps by negative SDSI values.

SDSI significance was assessed using a permutation test. For each neuron and turn direction, all trials with right sweeps and all trials with left sweeps were pooled together and randomly reassigned into two groups whose sizes matched the original sample sizes. SDSI was then recomputed. This procedure was repeated 1000 times to generate a shuffled distribution of SDSI values. For cells that fired more during trials with right sweeps, the p-value was defined as the fraction of shuffled SDSI values that were greater than or equal to the observed SDSI, or less than or equal to the observed SDSI for cells that fired more during trials with left sweeps. Thus, each cell was assigned an SDSI value and an associated p-value for each turn direction. SDSI values were deemed significant if their associated p-value was less than 0.05.

Of the turn cells with a significant SDSI value in one or both turn directions (38%, 64/167), the vast majority were selective in only one turn direction (88%, 56/64). We restricted subsequent analyses to this majority population as each cell had a single significant SDSI value that could be directly compared to its TDSI (Figure 4E). Across this population, most cells fired more during trials with sweeps in their preferred turn direction compared to trials with sweeps in their non-preferred turn direction ('sweep-selective turn cells', 75%, 42/56, bottom left and top right quadrants of Figure 4E). Of these, most cells (62%, 26/42) exhibited sweep direction selective firing during trials in which the animal turned in the cell's preferred direction (Figure 4C and 4D, Figure S6B and S6C) while the rest exhibited selectivity when the animal turned in the cell's non-preferred direction (Figure S6D). Together, these results reveal that turn cell activity is modulated in coordination with sweeps primarily during trials in which the animal turns in the cell's preferred direction.

Sweep-selective turn cell activity relative to trials without sweeps

To determine whether the activity of sweep-selective dSC neurons can be bidirectionally modulated, that is increase or decrease their firing in coordination with sweep presence and direction, we analyzed the firing rates of sweep-selective turn cells (n = 42) around the time of the bifurcation across three distinct trial types: trials with sweeps 1) only in the cell's preferred direction, 2) only in the cell's non-preferred direction, and 3) in both directions, all relative to trials without sweeps.

For each sweep-selective turn cell, we calculated a normalized firing rate modulation score for each of the three distinct trial types above as the difference between the average spike count during that trial type and the average spike count during trials without sweeps, divided by their sum $((\text{trial type spike count} - \text{no sweep spike count}) / (\text{trial type spike count} + \text{no sweep spike count}))$, calculated over the same 200 ms bifurcation window used for SDSI. Accordingly, each sweep-selective turn cell was assigned a modulation score for each trial type (Figure 4F). Positive values indicate that a cell fired more on average during that trial type compared to no sweep trials, negative if it fires less, and near zero if there is no change.

Generalized linear models for assessing if hippocampal sweeps predict turn cell firing

We built Poisson family generalized linear models (GLMs, log link function) to determine whether and when hippocampal sweeps prior to the bifurcation could predict dSC activity at the bifurcation across trials (Figure 5). We included trials with sweeps exclusively in one direction (ipsi or contra) as well as trials without sweeps, excluding trials with sweeps in both directions.

We divided each trial into theta cycles prior to and at the time of the bifurcation (cycles -4 through -1 and cycle 0, respectively) and, for each cycle, extracted three binary predictors across trials: turn direction (left = 1, right = 0), left sweep (1 or 0), and right sweep (1 or 0). For each sweep-selective turn cell ($n = 42$), the response variable was defined as the spike count across trials during the bifurcation cycle (cycle 0). Importantly, separate GLMs were fit for each cycle (-4 through 0), with predictors drawn from that cycle and the response variable fixed as turn cell spike counts at cycle 0.

For each turn cell, we extracted the beta coefficients for the left and right sweep predictors from each GLM, resulting in 10 beta coefficients per cell (2 sweep directions x 5 theta cycles). The significance of each coefficient was assessed using a Wald z-test. Significant beta coefficients using a conservative threshold ($p < 0.01$) were converted to their sign (± 1), indicating whether sweeps during a given theta cycle significantly predicted increased firing (+1, green-colored cycles in Figure 5) or decreased firing (-1, purple-colored cycles in Figure 5) at the bifurcation cycle. Such cycles were deemed 'predictive cycles.'

We performed permutation tests (1000 iterations) to assess whether the observed proportion of turn cells with predictive cycles and the observed proportion of total predictive cycles exceeded chance. For each iteration, spike counts for each turn cell were shuffled separately within left and right turn trials to disrupt sweep-firing relationships. P-values for cells and cycles were calculated as the proportion of shuffles in which the shuffled statistic met or exceeded the observed values. Both the proportion of turn cells with predictive cycles and the proportion of predictive cycles significantly exceeded shuffled controls (Figure S7A).

To assess whether adding sweep predictors improved model performance for each significantly predictive cycle ($n = 32$ total cycles, Figure 5B), we compared a full model (turn direction predictors + sweep predictors) to a reduced model (turn direction predictors only) using 5-fold cross-validation. Model improvement was quantified as percentage reduction in mean absolute error (MAE) between the reduced model and the full model calculated as: $(\text{MAE reduced model} - \text{MAE full model}) / \text{MAE reduced model} \times 100$. The observed positive values averaged across folds indicate that adding sweep predictors improved model prediction beyond turn direction alone (Figure S7B).

Generalized linear models for assessing if turn cell firing predicts hippocampal sweeps

We built binomial family GLMs (logit link function) to test whether and when dSC turn cell firing prior to the bifurcation could predict the direction of hippocampal sweeps at the bifurcation across trials (Figure S8). For this analysis, we included trials with sweeps exclusively in one direction (ipsi or contra), excluding trials with sweeps in both directions as well as trials without sweeps.

As in the previous GLM analysis, we divided each trial into theta cycles spanning cycles -4 through 0 relative to the bifurcation. For each cycle, we extracted a binary turn-direction predictor (left turn = 1, right turn = 0) and spike counts for a given sweep selective turn cell across trials. For the response variable, we identified whether a hippocampal sweep occurred in cycle 0, and, if so, assigned sweep direction as left = 1 and right = 0. Importantly, separate GLMs were fit for each cycle (-4 through 0), with predictors drawn from that cycle and the response variable fixed as sweeps at cycle 0.

For each sweep-selective turn cell ($n = 42$), we extracted the beta coefficients for the spike-count predictors from cycles -4 through 0 in each GLM. The significance of each coefficient was assessed using a Wald z-test. Significant coefficients ($p < 0.01$) were converted to their sign (± 1), indicating whether higher spike counts during a given theta cycle significantly predicted leftward sweeps (+1, blue-colored cycles in Figure S8) or rightward sweeps (-1, red-colored cycles in Figure S8) at the bifurcation. Interestingly, at the $p < 0.01$ level as in the previous GLM analysis (Figure 5), we observed only a single cycle in which dSC turn cell activity predicted sweeps at the bifurcation, and this cycle involved activity from a left turn cell predicting rightward sweeps (Figure S8A). Implementing a less conservative significance threshold ($p < 0.05$), we did observe an increased proportion of cells and cycles in which dSC turn cell activity predicted future sweeps in a directionally consistent manner (i.e., left turn cells predicting left sweeps, right turn cells predicting right sweeps, Figure S8B); however, after performing permutation tests (1000 iterations, as described in the previous GLM analysis), we found that these proportions did not reliably exceed shuffled controls (Figure S8C).

Quantifying deviations in behavioral trajectory around the turn

We used two approaches to compare animals' behavioral trajectories around the bifurcation between ipsi and contra sweep trials.

The first involved quantifying trajectory divergence from a linear path. To obtain a linear path, we first calculated the average trajectories during ipsi and contra sweep trials for turns a given direction. We did so by interpolating each trial to the same number of samples and rotating them to a common coordinate frame (Figure 6A). We then defined a 'band of interest' starting with an upper boundary near the bifurcation point of the maze and a lower boundary extending 10 cm from the upper boundary toward the base of the maze. This band was chosen to focus on trajectory deviations immediately prior to the bifurcation. The linear path was defined by connecting the average positions at which the ipsi and contra sweep trajectories crossed the upper and lower boundaries of the band, producing a single linear path against which trajectory deviations were measured. We then calculated, within this band of interest, the deviation of each ipsi and contra sweep trial from the linear path as the average perpendicular distance from each sample point to the line, producing a single deviation value per trial. For the ROC analysis, deviation values for ipsi and contra sweep trials were treated as two distributions, and an ROC curve was computed. The optimal classification threshold was found by minimizing the distance to the top-left corner of the ROC curve. We calculated the balanced classification accuracy (% accuracy, $(\text{true positive rate} + \text{true negative rate}) / 2$) to assess ROC performance. For each

session, classification accuracy was calculated separately for left and right turn trials (Figure 6B, left). For each animal, classification accuracies were averaged across turn directions and sessions (Figure 6B, right).

Second, trajectory deviations were quantified using the IdPhi measure (the integrated absolute angular velocity)²⁸ calculated over the animal's trajectory (Figure 6C). Briefly, average trajectories for ipsi and contra sweep trials were first smoothed (Gaussian, 1 sample) and heading angle was derived. IdPhi was defined as the sum of the absolute change in heading velocity across each average trajectory, with higher values indicating greater deviations in heading. For each session, IdPhi was averaged across left and right turn trials (Figure 6C).

Quantification and statistical analysis

All analyses and statistical tests were implemented using custom Python scripts (Python version 3.8.5). Statistical tests used and p-values are provided throughout the text and in figure legends.

Acknowledgements. We thank P. Saraf, L. Ruan and Q.Y. Wu for technical support and the members of the Scanziani laboratory for helpful discussions of this project. This project was supported by the Neuroscience Training Grant T32 (C.W.), by the Discovery Fellowship of the UCSF Graduate Division (C.W.) and the Howard Hughes Medical Institute (M.S. and L.M.F.).

Author contributions. C.W., L.M.F. and M.S. designed the study. C.W. conducted all experiments and experimental data analysis. C.W., L.M.F. and M.S. wrote the manuscript.

Main Figures

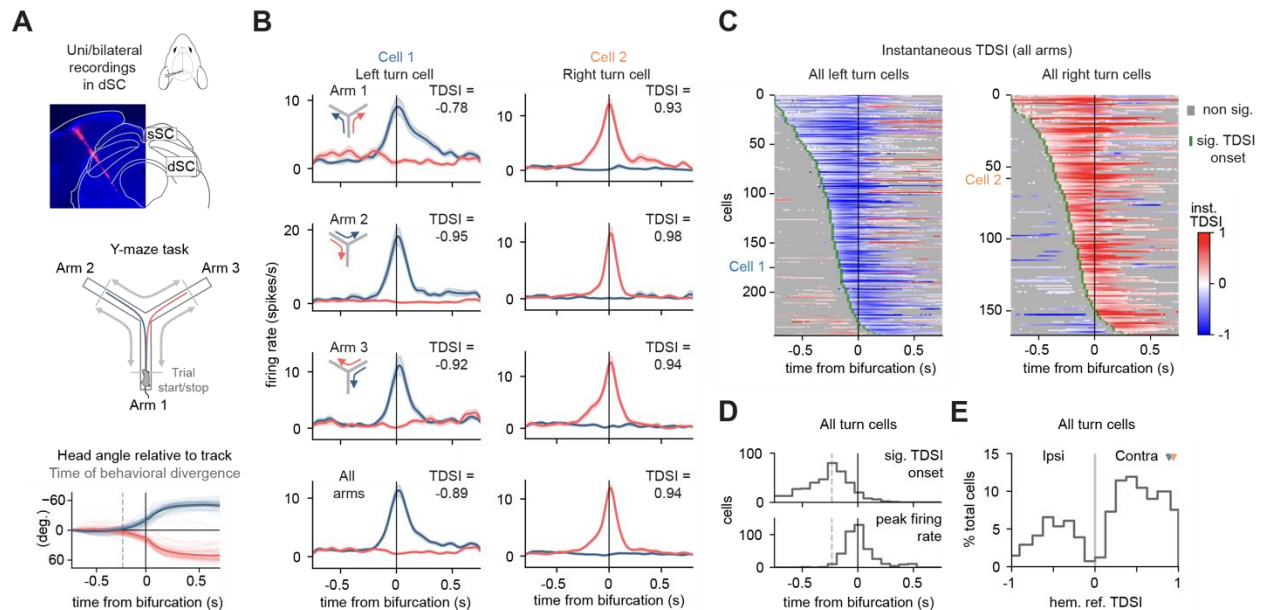


Figure 1: Turn cells in the intermediate and deep layers of the SC

(A) Recording configuration and Y-maze task. Top, schematic and histology of an example implantation targeting the intermediate and deep layers of the SC (dSC). Probe track is marked by Dil (pink). sSC = superficial layers of the SC. Middle, Y-maze with behavioral trajectories. Bottom, angle of the animals' head relative to the track during the task. Dark lines: average over all sessions and animals (n = 55 sessions, 12 mice); light lines: individual sessions. Dashed line: time from the maze bifurcation at which the average head angle relative to the track significantly diverges between right and left turns (-0.25 s).

(B) Firing rates of two example turn cells for all possible trajectories on the Y-maze. Left, example left turn cell (Cell 1). Right, example right turn cell (Cell 2). Blue and red are left and right turn trajectories, respectively. Bottom panels are the averages of the firing rates across all left or right turn trajectories. Thick lines: average firing rates across trials, shaded bands represent s.e.m. Note negative and positive TDSI (Turn Direction Selectivity Index) values for Cell 1 and 2 respectively, across all arms (TDSI: p < 0.001 for each arm, permutation tests).

(C) All left and right turn cells sorted by the time at which the instantaneous TDSI became selective in the cell's preferred turn direction (significant TDSI onset; n = 244 left turn cells, n = 167 right turn cells; 8 right dSC insertions, 34 sessions; 6 left dSC insertions, 28 sessions).

(D) Distributions of significant TDSI onset times and times of peak firing rate across all turn cells. Dashed lines: average time of behavioral divergence as shown in A.

(E) Distribution of TDSI values across all turn cells referenced to the recorded dSC hemisphere. Positive and negative hemisphere referenced TDSI values indicate turn cells whose firing is selective for contraversive (e.g. left turn cells in the right dSC) or ipsiversive (e.g. right turn cell in right dSC) turns, respectively. Markers indicate hemisphere referenced TDSI values for Cell 1 and Cell 2.

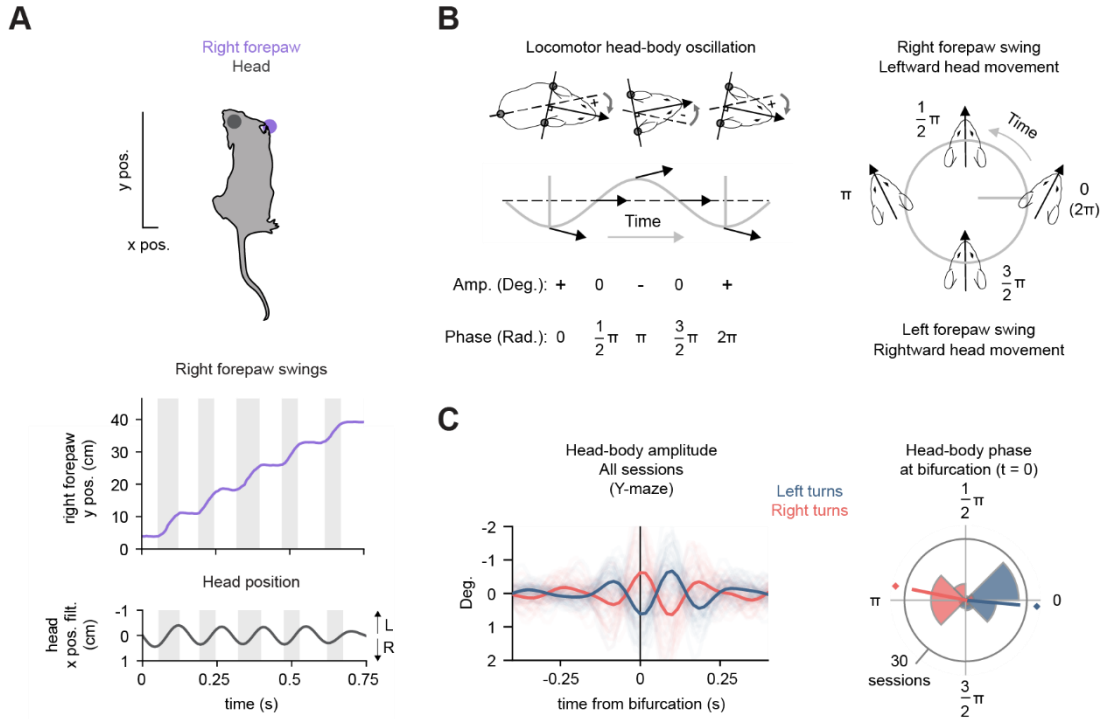


Figure 2: Turns are coordinated with the stepping cycle in mice

(A) Above, schematic of mouse with labels on the right forepaw and head (as labeled on frames taken with the bottom view camera; scale bar: 1 cm x, 8 cm y). Below, example pass on linear track. Top panel: Right forepaw position along the y-axis (y pos.). Gray bands indicate times of right forepaw swings. Bottom panel: Head position along the x-axis (x pos.) filtered between 4-10 Hz. Note coupling between right forepaw swings and leftward head movements.

(B) Left, schematic illustration of head-body angle during locomotion. Positive and negative amplitude values reflect right and left head direction relative to the body, respectively. The phase of the head-body oscillation cycle is given in radians. Right, polar plot showing head and forepaw movements associated with each quarter phase of the head-body oscillation cycle.

(C) Left, amplitude of head-body angle filtered between 4-8 Hz around the time animals reached the bifurcation across all sessions color coded by turn direction. Time from bifurcation was defined as the time at which the animal's head position crossed the center point of the maze. Dark lines: average over all sessions and animals ($n = 55$ sessions, 12 mice); light lines: individual sessions. Note similar amplitude yet opposite sign of the head-body angle around the bifurcation. Right, distributions of instantaneous head-body phase at the time the animals reached the bifurcation ($t = 0$) for each turn direction across sessions. Left turns: $p < 0.001$ Rayleigh test, right turns: $p < 0.001$, Rayleigh test. Paired Rayleigh test between left and right turns: $p < 0.05$.

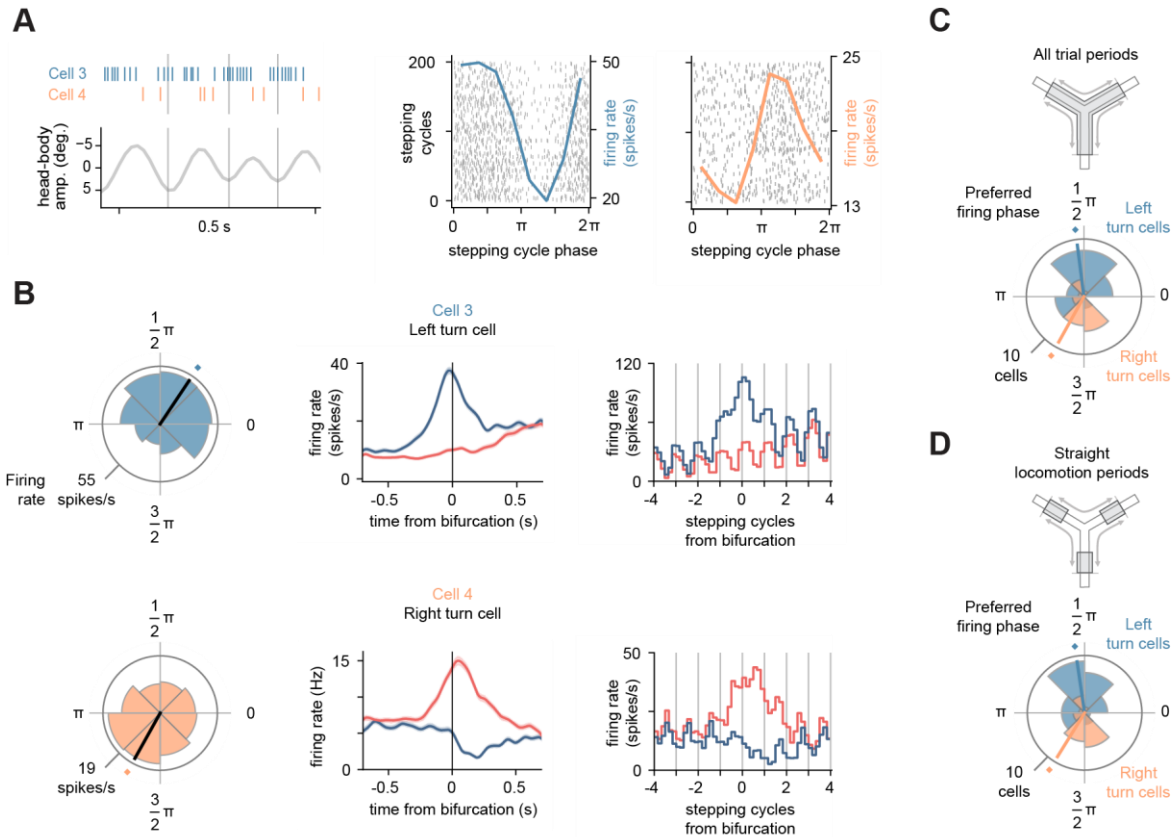


Figure 3: Turn cell activity is coupled to the stepping rhythm

(A) Left, spike raster of two simultaneously recorded turn cells, Cell 3 and Cell 4 (top) and ongoing head-body oscillation (bottom). Vertical gray lines indicate phase 0 (2π). Middle and right, spike raster plots of Cell 3 and Cell 4, respectively, aligned to the first 200 head-body oscillation cycles in the session. The overlaid traces are the firing rates averaged over all head-body oscillation cycles in the session (right y-axis). Note that the firing of these two cells is coupled to opposite phases of the head-body oscillation cycle.

(B) Left, polar plots of the firing of Cell 3 and Cell 4 relative to the phase of the head-body oscillation cycle for the session shown in A. Cell 3: $\ln(Z)$: 7.15, $p < 0.001$, Rayleigh test, preferred firing phase: 0.98 radians; Cell 4: $\ln(Z)$: 5.30, $p < 0.001$ Rayleigh test, preferred firing phase: 4.20 radians. Middle, firing rates of Cell 3 and Cell 4 around the time of left (blue) and right (red) turns showing their opposite selectivity for turn direction (Cell 3: TDSI = -0.55; Cell 4: TDSI = 0.56). Right, firing rates of Cell 3 and Cell 4 aligned to head-body cycles from turn. Vertical gray lines indicate phase 0 (2π).

(C) Preferred firing phases of left and right turn cells during all trial periods (left turn cells (TDSI < -0.1), population: $p < 0.01$, right turn cells (TDSI > 0.1), population: $p < 0.05$, Rayleigh tests, left vs. right turn cell populations: $p < 0.05$, permutation test).

(D) Preferred firing phases of left and right turn cells during straight locomotion periods only (left turn cells, population: $p < 0.01$, right turn cells, population: $p < 0.05$, Rayleigh tests, left vs. right turn cell populations: $p < 0.05$, permutation test). Note that cells maintain the same preferred firing phase during turns and straight locomotion.

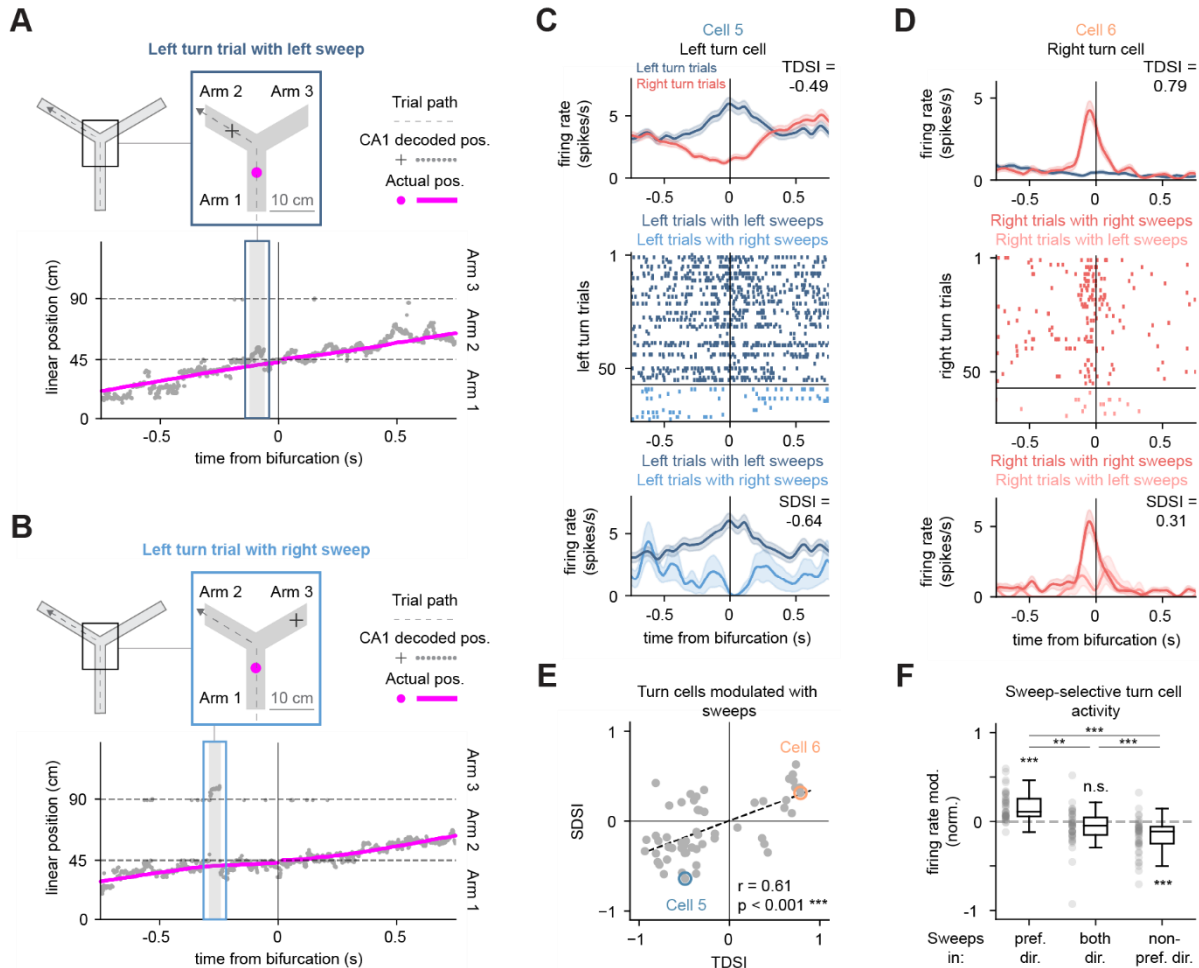


Figure 4: Turn cell activity is modulated in coordination with hippocampal sweeps

(A) Example left turn trial with left hippocampal sweep. Top (expanded view), maze schematic showing a snapshot of the animal's actual position (pink circle) and decoded position (black crosshair) during the sweep. The position of each marker reflects the average linearized position, either actual or decoded, across the time window indicated by the outlined gray band in the bottom plot. Bottom, plot showing the decoded position (gray) and actual position (pink) relative to the maze bifurcation (y-axis, linearized position). Note the decoded position sweeping up into the left arm as the animal approaches the bifurcation.

(B) Same as in A, but for a left turn trial with a right hippocampal sweep during the same session.

(C) Top, firing rate profiles of a left turn cell in the dSC (Cell 5). Data taken from the same session as in A and B. Middle, spike raster plot for Cell 5 during left turn trials. Dark blue spikes indicate spikes during left turn trials with left sweeps; light blue spikes indicate spikes during left turn trials with right sweeps. Bottom, firing rate profiles of Cell 5 during left turn trials separated by sweep direction (same color scheme as in middle panel). Note increased firing during left turn trials with left sweeps compared to left turn trials with right sweeps corresponding to a significantly negative Sweep Direction Selectivity Index value (SDSI: $p < 0.01$, permutation test).

(D) Same as in C but for a right turn cell (Cell 6) during right turn trials. Data taken from a different session than in A-C. Note increased firing during right turn trials with right sweeps compared to right turn trials with left sweeps corresponding to a significantly positive SDSI value ($p < 0.05$, permutation test).

(E) Correlation between TDSI and SDSI across all cells whose activity was significantly modulated with sweep direction ($n = 56$ cells, Pearson $r = 0.61$, $p < 0.001$).

(F) Firing rate modulation of sweep-selective turn cells during trials with sweeps only in the cell's preferred direction, non-preferred direction, or both directions, normalized by the cell's firing during trials without sweeps. Note bidirectional modulation of firing compared to the no sweep condition (preferred/non-preferred only: $p < 0.001$, both: $p > 0.05$, 1-sample t-tests, Bonferroni-corrected). These distributions were significantly different from one another (pref. vs both: $p < 0.01$, pref. vs non-pref: $p < 0.001$, and both vs non-pref: $p < 0.001$, paired t-tests, Bonferroni-corrected).

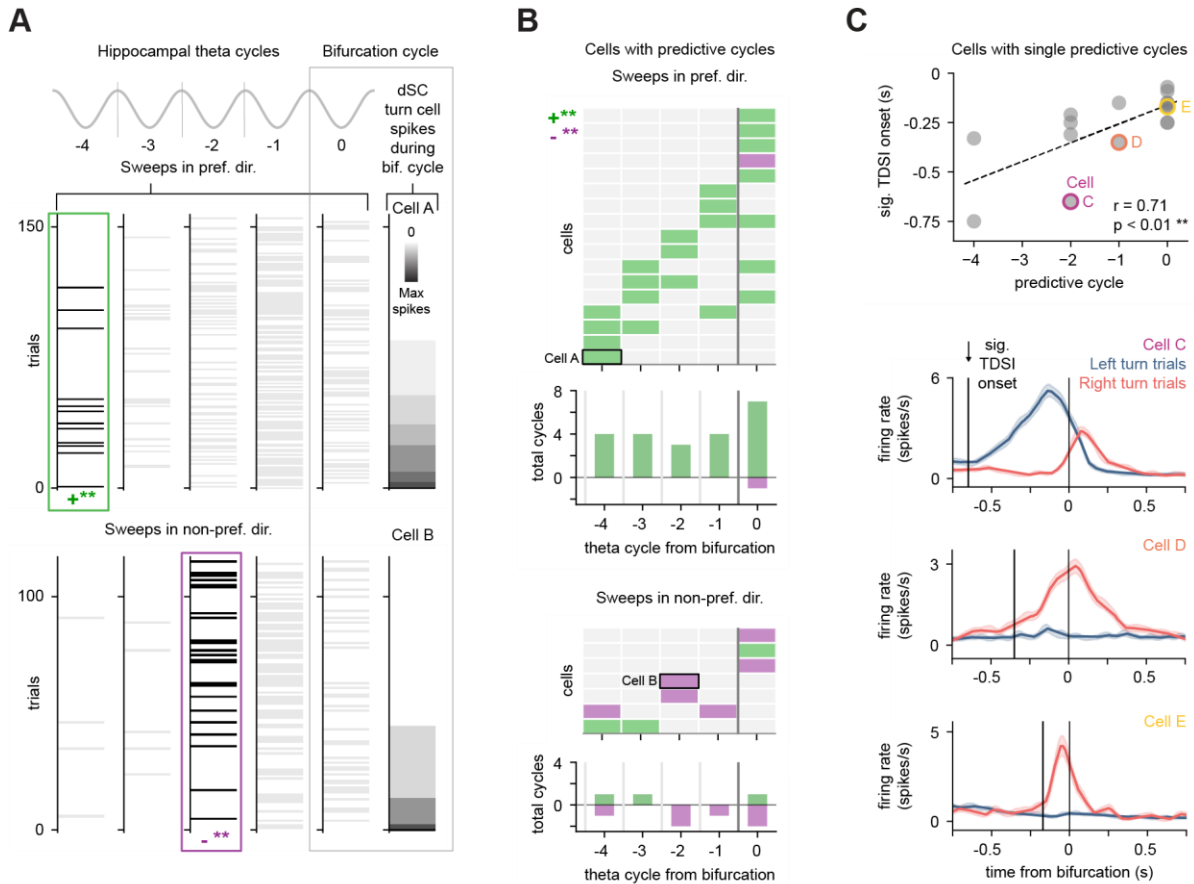


Figure 5: Hippocampal sweeps reliably predict future turn cell activity

(A) GLM analysis to assess whether the presence and direction of hippocampal sweeps during individual theta cycles prior to and at the time of the bifurcation (cycles -4 through -1 and cycle 0, respectively) predicts dSC turn cell firing at the bifurcation. Top, GLM analysis for an example dSC turn cell (Cell A). Spike count for Cell A during the bifurcation cycle (cycle 0) is shown at the far right. Trials are sorted by increasing spike counts of Cell A. Hippocampal sweeps in Cell A's preferred turn direction across trials are plotted as horizontal lines during their corresponding theta cycle relative to the bifurcation (cycles -4 through 0). Trials are those in which the animal turns in the cell's preferred turn direction. Note that sweeps occurring during cycle -4 are associated with increases in spiking of Cell A during the bifurcation cycle. Significantly positive (+**) predictive cycle is shown by a green outline ($p < 0.01$, Wald z-tests). Bottom, GLM analysis for another example turn cell (Cell B) in which sweeps in Cell B's non-preferred turn direction are shown. Note that sweeps occurring during cycle -2 across trials are associated with decreases in spiking of Cell B during the bifurcation cycle. Significantly negative (-**) predictive cycle is shown by a purple outline ($p < 0.01$, Wald z-tests). Sweeps occurring during significantly predictive cycles are shown in black to highlight the relationship between sweeps and turn cell firing. Sweeps occurring during non-significantly predictive cycles are shown in gray.

(B) Top, all cells with predictive cycles for sweeps in the cell's preferred turn direction sorted by theta cycle relative to the bifurcation. Bottom, all cells with predictive cycles for sweeps in the cell's non-preferred turn direction.

(C) Top, correlation between timing of predictive cycle and onset of significant TDSI across turn cells. This analysis was necessarily done only on cells whose activity was either positively or negatively predicted by sweeps during a single theta cycle ($n = 16$ cells, Pearson $r = 0.71$, $p < 0.01$). Bottom, firing rate profiles of three example turn cells (Cell C, D, and E from top plot) showing progressively later significant TDSI onsets relative to the turn.

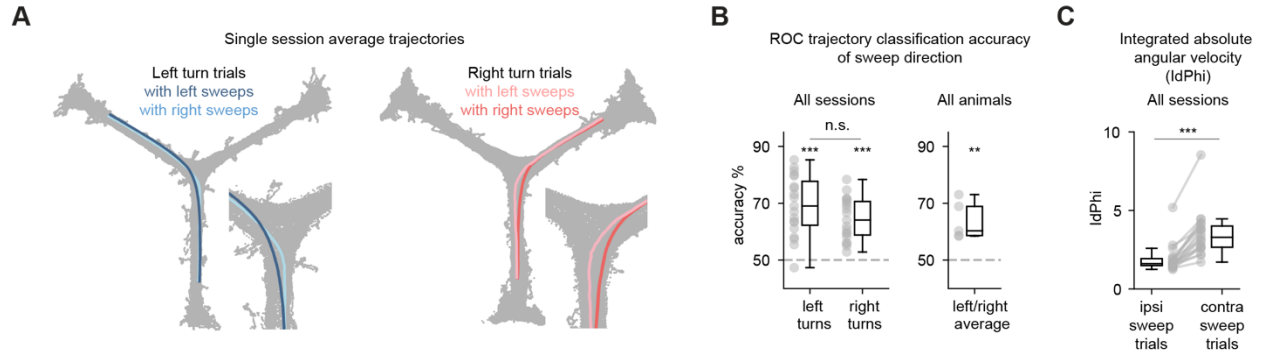


Figure 6: Hippocampal sweep direction and deviations in trajectory

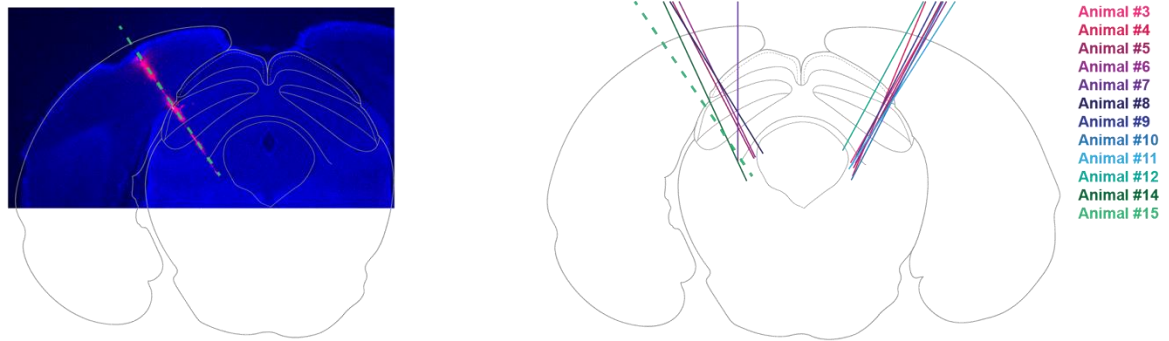
(A) Left, average left turn trajectories for a single session color coded by sweep direction. Dark blue trajectory indicates average of left turn trials with left sweeps (ipsi sweep trials), light blue trajectory indicates average of left turn trials with right sweeps (contra sweep trials). Session is the same session as in Figure 4A, 4B, and 4C. Note deviation of trajectory for contra sweep trials toward the alternative path. Right, same as in A but for right turn trials. Session is the same session as in Figure 4D.

(B) Left, ROC classification accuracy of sweep direction from deviation in behavioral trajectory from a linear path around the time of turns (see Methods), shown across sessions for both turn directions ($n = 19$ sessions, left turns: $69\% \pm 2\%$, right turns: $64\% \pm 2\%$, left and right turns: $p < 0.001$, 1-sample t-tests compared to 50%, Bonferroni-corrected; left vs. right turns: $p > 0.05$, paired t-test). Right, ROC classification accuracy between ipsi and contra sweep trials, combined across left and right turns per animal ($n = 5$ animals, $64\% \pm 3\%$, $p < 0.01$, 1-sample t-test compared to 50%).

(C) Integrated absolute angular velocity (IdPhi) values calculated across the average trajectories for ipsi vs. contra sweep trials for each session ($n = 19$ sessions, $p < 0.001$, paired t-test). Values are averaged between left and right turns.

Supplemental Figures

A



B

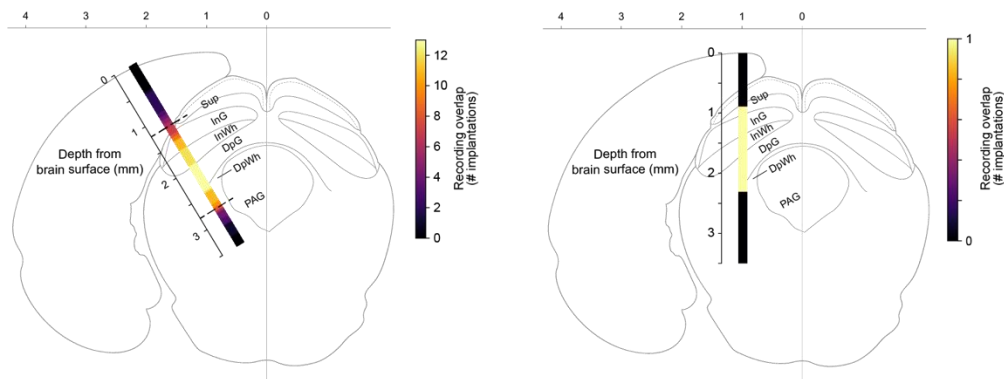


Figure S1: Histology and recording site analysis (dSC)

(A) Left, example coronal section of an implantation targeting the dSC (animal #15). Probe track is marked by Dil (pink) and indicated by a dashed green line. Same as in Figure 1A. Right, probe tracks from all implantations ($n = 14$ implantations from 12 mice (10 unilateral, 2 bilateral)).

(B) Left, heatmap showing the approximate span of recording depths in the SC across implantations in which the probe was inserted at an angle. Lighter colors indicate higher overlap. Dashed black lines indicate the average depths from the brain surface of the first and last dSC recordings. Note high concentration of recordings between the intermediate gray (InG) and deep white layers (DpWh). Heatmap is oriented at the average implantation angle and average ML/AP coordinates across animals (31° , ML: 2.25 mm, AP: 0 mm from lambda (-4.04 mm from bregma)). Right, heatmap showing the approximate span of recording depths for the vertical implantation. (Sup = superficial layer, InG = intermediate gray layer, InW = intermediate white layer, DpG = deep gray layer, DpWh = deep white layer, PAG = periaqueductal gray, adapted from The Mouse Brain in Stereotaxic Coordinates⁶⁹).

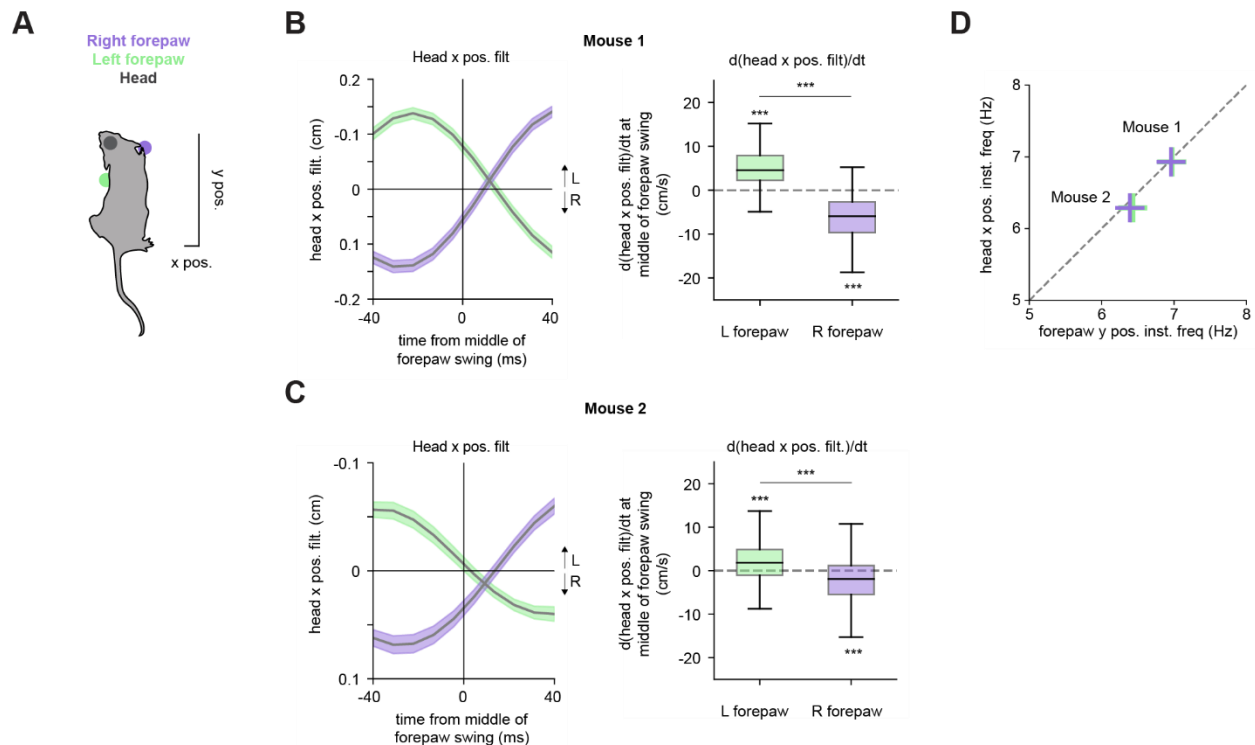


Figure S2: Relationship between stepping and left-right head movements in mice

(A) Schematic of mouse with labels on the right forepaw, left forepaw, and head (as labeled on frames from the bottom view camera).

(B) Left, head position along the x-axis (filtered between 4-10 Hz) for an example mouse (Mouse 1) aligned to the middle of right or left forepaw swings (color code as in (A) thick lines indicate average, shaded bands indicate s.e.m., $n = 114$ right forepaw swings, 105 left forepaw swings that met criteria (see Methods)). Right, rate of change of head position at the middle of each forepaw swing. Note positive (rightward) rate of change during left forepaw swings and negative (leftward) rate of change during right forepaw swings ($p < 0.001$ for both left and right forepaws, 1-sample t-tests compared to 0, left vs. right: $p < 0.001$, 2-sample t-test).

(C) Same as in (B) but for another example animal (Mouse 2, $n = 293$ right forepaw swings, 326 left forepaw swings, $p < 0.001$ for both left and right forepaws, 1-sample t-tests compared to 0, left vs. right: $p < 0.001$, 2-sample t-test).

(D) Average forepaw oscillation frequency in y position compared to average head oscillation frequency in x position. Both signals were filtered between 4 and 10 Hz.

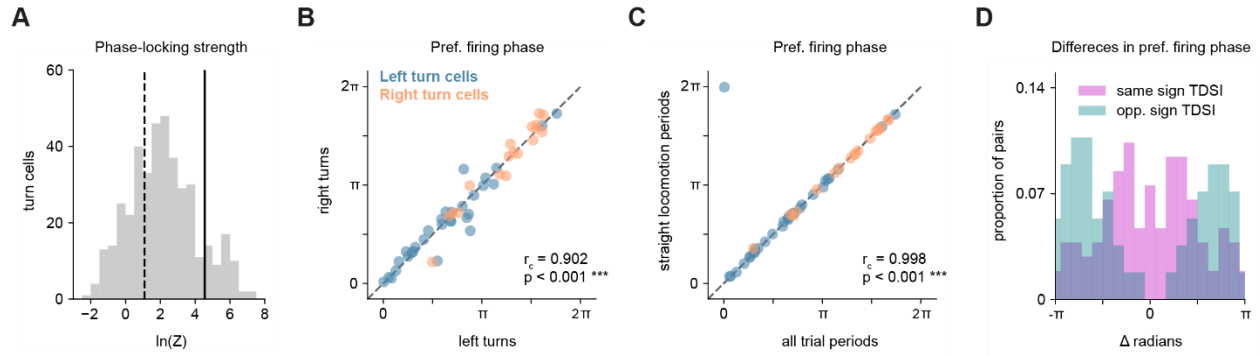


Figure S3: Quantification of phase-locking and phase relationships in rhythmically active turn cells

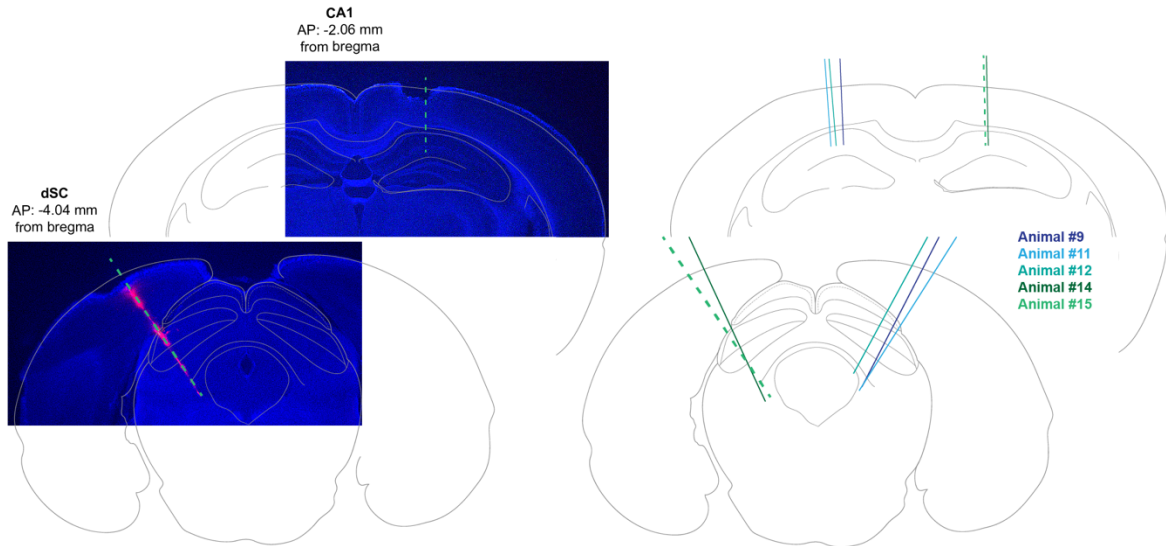
(A) Distribution of log-transformed Rayleigh's Z values across all turn cells ($n = 411$ cells). $\ln(Z)$ values to the right of the dashed vertical line are significant at $p < 0.05$ ($n = 282$ cells). $\ln(Z)$ values to the right of the solid vertical line indicate turn cells with the strongest rhythmic activity ($n = 49$ cells; solid vertical line: mean + SD of $\ln(Z)$ values that are significant at the $p < 0.05$ level).

(B) Correlation between preferred firing phases of left and right turn cells during left vs. right turn trials ($r_c = 0.902$, $p < 0.001$, permutation test). Same populations as in Figure 3C.

(C) Correlation between preferred firing phases of left and right turn cells during all trial periods vs. straight locomotor periods ($r_c = 0.998$, $p < 0.001$, permutation test). Same populations as Figure 3C.

(D) Differences in preferred firing phases between simultaneously recorded cell pairs color coded by whether the pair had TDSI values with the same or opposite sign (same sign pairs (i.e., left turn cell vs. left turn cell), mean angle: 0 ± 0.17 radians; opposite sign pairs (i.e., left turn cell vs. right turn cell), mean angle: -3.06 ± 0.18 radians). Note clustering of same sign pairs around 0 and clustering of opposite sign pairs around $\pm \pi$ (same sign pairs: $p < 0.01$, Rayleigh test; opposite sign pairs: $p < 0.001$, Rayleigh test).

A



B

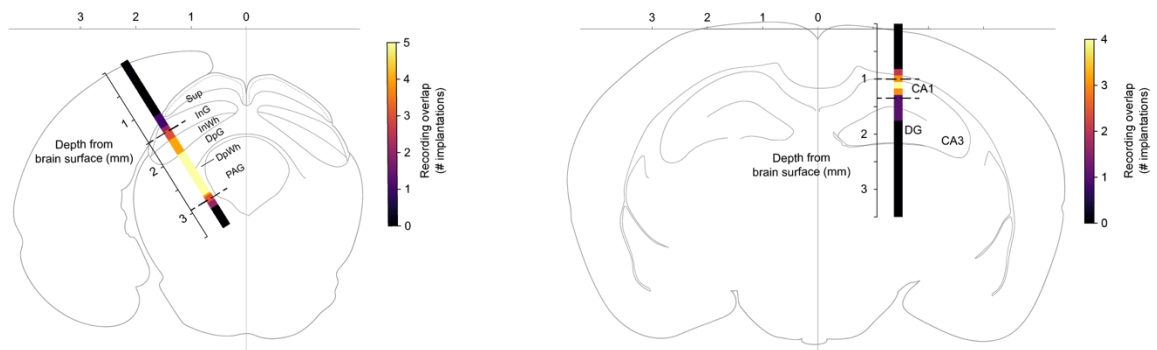


Figure S4: Histology and recording site analysis (dSC and CA1)

(A) Left, example coronal sections of probe implantations targeting the dSC and dorsal CA1 regions in an example animal (animal #15). Probe track in dSC is marked by Dil (pink) and indicated by a dashed green line. dSC probe tracks are a subset of those in Figure S1. Probe track in CA1 is indicated by dashed green line. Right, probe tracks in dSC and dorsal CA1 across all implantations (n = 5 implantations from 5 mice).

(B) Left, heatmap showing the approximate span of recording depths in the SC across implantations. Lighter colors indicate higher overlap. Dashed black lines indicate the average depths from the brain surface of the first and last dSC recordings. Note high concentration of recordings between the intermediate gray (InG) and deep white layers (DpWh). Heatmap is oriented at the average implantation angle and ML/AP coordinates across animals. (Sup = superficial layer, InG = intermediate gray layer, InW = intermediate white layer, DpG = deep gray layer, DpWh = deep white layer, PAG = periaqueductal gray). Right, same as left panel but for CA1 implantations. All outlines of histological sections were adapted from The Mouse Brain in Stereotaxic Coordinates⁶⁹.

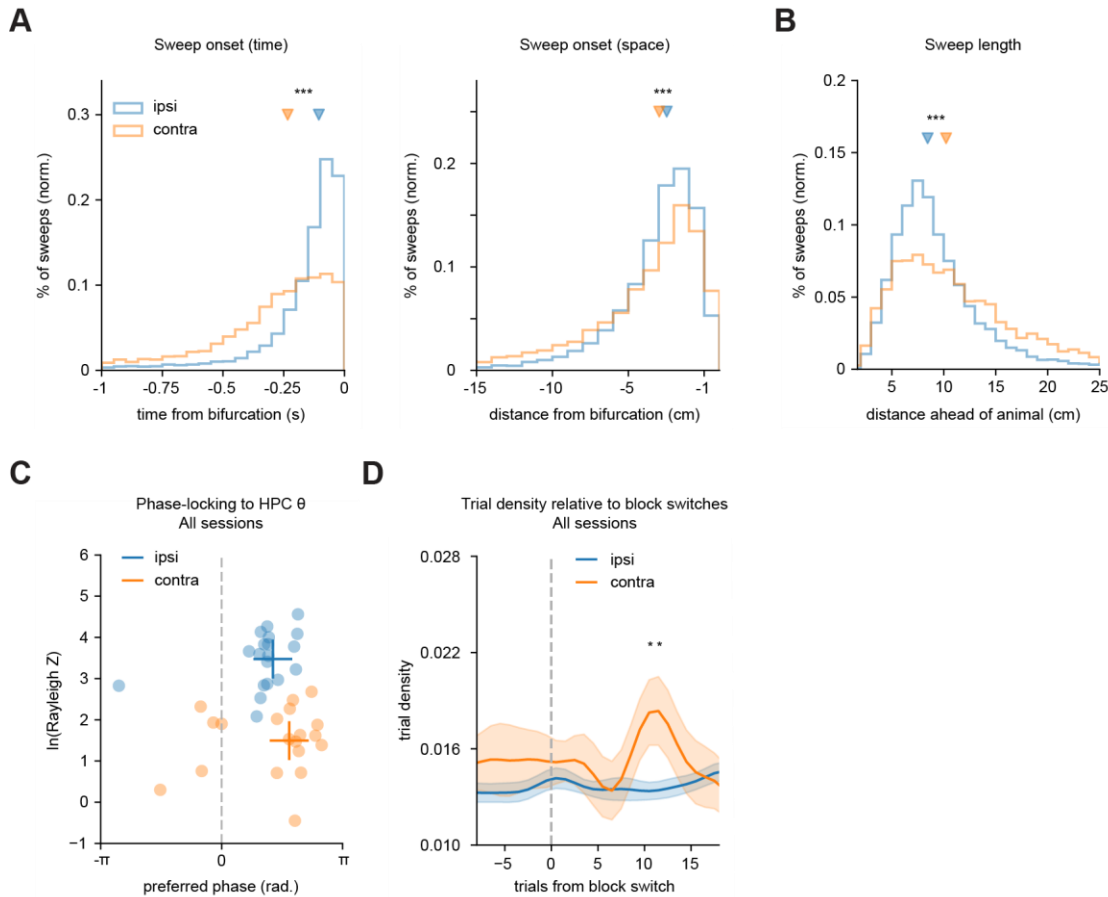


Figure S5: Ipsi and contra sweep parameters

(A) Left, distributions of sweep onset times relative to the turn for ipsi and contra sweeps (ipsi sweeps (sweeps in same direction of turn), blue, $n = 9178$ total sweeps, median: -105 ms, IQR: -54 to -201 ms, contra sweeps (sweeps in opposite direction of turn), orange, $n = 3002$ total sweeps, median: -234 ms, IQR: -115 to -403 ms, $p < 0.001$, Mann-Whitney U test). Right, distributions of the animals' distance from the turn at the onset of ipsi and contra sweeps (ipsi sweeps, median: -3.47 cm, IQR: -2.21 to -5.40 cm, contra sweeps, median: -3.96 cm, IQR: -2.20 to -7.31 cm, $p < 0.001$, Mann-Whitney U test).

(B) Distributions of sweep distance from the animal's position for ipsi and contra sweeps (ipsi sweeps, median: 8.45 cm, IQR: 6.47-11.29 cm, contra sweeps, median: 10.23 cm, IQR: 6.76-14.93 cm, $p < 0.001$, Mann-Whitney U test). Sweep distance was calculated as the average difference between the animal's actual and decoded positions during the duration of the sweep.

(C) Phase-locking of sweeps to late phases (0 to π rad.) of the hippocampal theta cycle across sessions ($n = 19$ sessions, ipsi, pref. phase: $+1.33 \pm 0.13$ rad., contra, pref. phase: $+1.76 \pm 0.26$ rad., ipsi vs. contra: $p > 0.05$, permutation test). X-axis indicates hippocampal theta phase where phase 0 corresponds to the trough of the theta cycle recorded at the CA1 pyramidal cell layer. Y-axis indicates strength of phase-locking. Note that, while both types of sweeps were phase-locked to late phases of theta, contra sweeps were more weakly phase-locked compared to ipsi sweeps (ipsi, $\ln(\text{Rayleigh } Z)$: 3.48 ± 0.15 , contra, $\ln(\text{Rayleigh } Z)$: 1.49 ± 0.18 , ipsi vs. contra: $p < 0.001$, paired t-test). Markers indicate average values.

(D) Density of trials containing only ipsi or contra sweeps aligned to block switches across sessions ($n = 19$ sessions). Traces were normalized by the number of trials of each type (ipsi or contra) per session and smoothed with a Gaussian kernel (2 bins). Thick lines represent the mean and shaded bands represent s.e.m. Note the transient increase in the density of contra sweep trials following block switches relative to ipsi sweep trials ($p < 0.05$ per bin, permutation tests), consistent with recent findings in rats showing an increased proportion of representations of alternative possibilities during periods of increased cognitive demand²⁹.

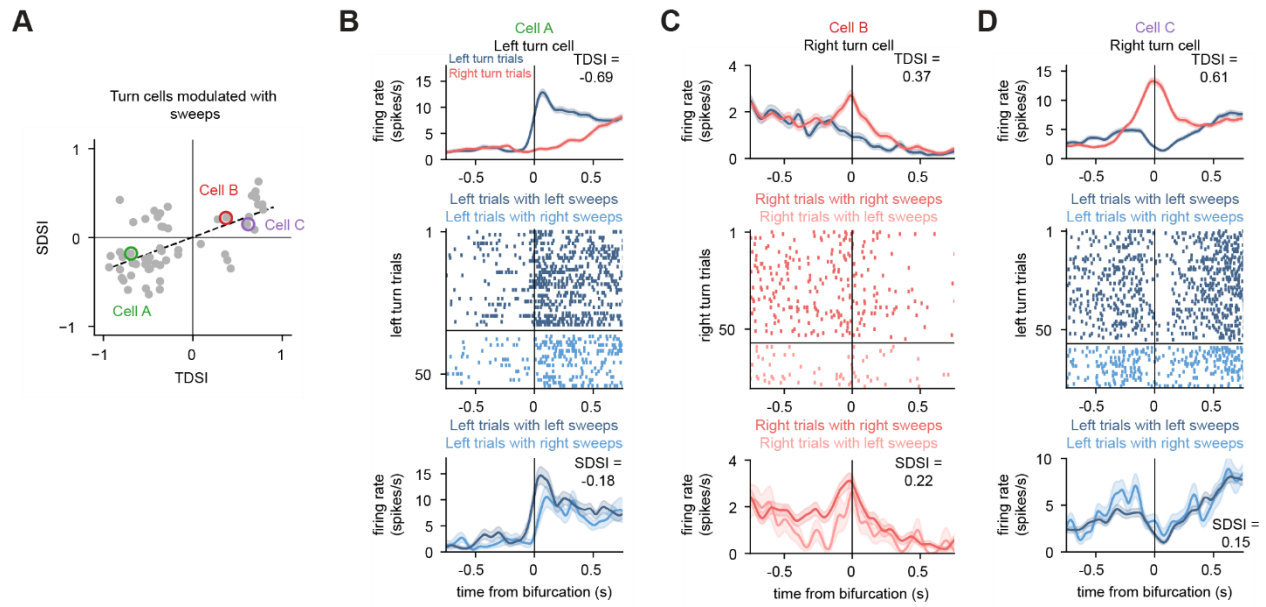


Figure S6: Additional examples of sweep-selective turn cells

(A) Same scatter plot as in Figure 4E. Markers indicate three additional sweep-selective turn cells shown in panels B-D.

(B) Example of left turn cell firing more during left turns with left sweeps compared to left turns with right sweeps ($p < 0.05$, permutation test).

(C) Example of right turn cell firing more during right turns with right sweeps compared to right turns with left sweeps ($p < 0.01$, permutation test).

(D) Example of right turn cell firing more during left turns with right sweeps compared to left turns with left sweeps ($p < 0.05$, permutation test).

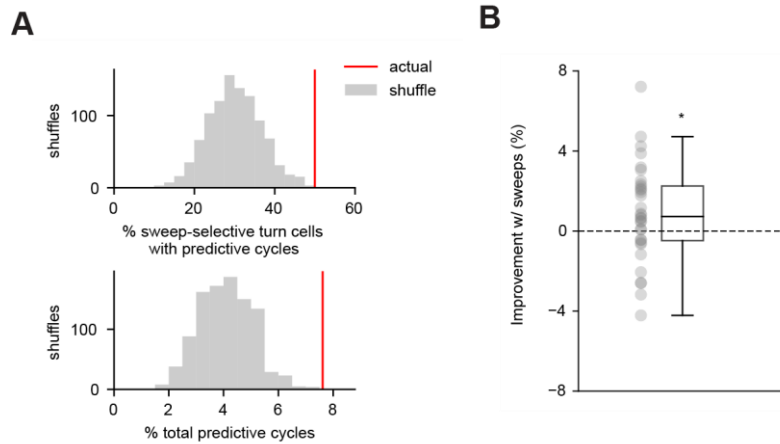


Figure S7: GLM shuffle controls and cross-validation for Figure 5

(A) Top, distribution of the percentage of sweep-selective turn cells with predictive cycles after shuffling spike counts across trials to disrupt sweep-firing relationships ($n = 1000$ shuffles, shuffled: $29\% \pm 0.2\%$, actual = 50% , $p = 0.001$: 1 shuffle had a percentage of sweep selective turn cells that exceeded that of the actual data). Bottom, same as top but for percentage of significantly predictive cycles ($n = 1000$ shuffles, shuffled: $4.1\% \pm 0.03\%$, actual = 7.6% , $p < 0.001$: 0 shuffles had a percentage of predictive cycles that exceeded that of the actual data).

(B) Model improvement with sweep information compared to turn direction information alone for each significantly predictive cycle ($n = 32$ cycles from Figure 5B, mean: $0.89 \pm 0.43\%$ (s.e.m.), 95% CI $[0.054, 1.72]\%$, $p < 0.05$, 1-sample t-test compared to 0). Model improvement is defined as percentage reduction in mean absolute error (MAE) when adding sweep predictors (full model) to a turn-only model (reduced model), calculated as: $(\text{MAE reduced model} - \text{MAE full model}) / \text{MAE reduced model} \times 100$. Positive values indicate model improvement with sweeps. For cross-validation, values were calculated and averaged across folds of a 5-fold train-test split.

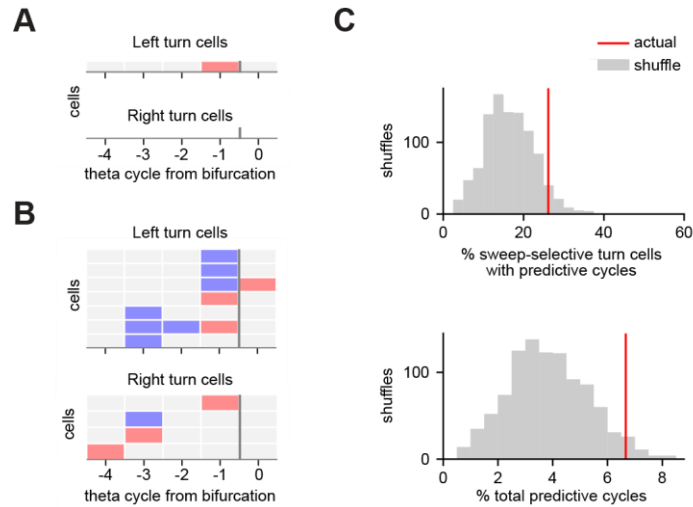


Figure S8: Turn cell activity does not reliably predict future hippocampal sweeps

(A) GLM analysis to assess whether dSC turn cell activity during individual theta cycles prior to and at the time of the bifurcation (cycles -4 through -1 and cycle 0, respectively) can predict the presence and direction of hippocampal sweeps at the bifurcation. Shown are all cells with predictive cycles at a $p < 0.01$ threshold (Wald z-tests). Shaded red boxes indicate cycles during which dSC firing predicted rightward sweeps at cycle 0. Shaded blue boxes, if present, indicate cycles during which dSC firing predicted leftward sweeps at cycle 0. Note that only activity for one left turn cell at cycle -1 predicted rightward sweeps at cycle 0, and no right turn cells exhibited predictive cycles.

(B) All left and right turn cells with predictive cycles at a $p < 0.05$ threshold (Wald z-tests) sorted by theta cycle relative to the bifurcation. Color code is same as in A. Note increased number of cells with predictive cycles (i.e. theta cycles during which dSC activity significantly predicted sweeps at the bifurcation) compared to A.

(C) Shuffle controls for cells and cycles in B ($p < 0.05$ threshold). Top, distribution of the percentage of sweep-selective turn cells with predictive cycles after shuffling spike counts across trials to disrupt firing-sweep relationships ($n = 1000$ shuffles, shuffled: $17 \pm 0.2\%$, actual = 26% , $p = 0.079$: 79 out of 1000 shuffles had a percentage of sweep selective turn cells that exceeded that of the actual data). Bottom, same as top but for percentage of significantly predictive cycles ($n = 1000$ shuffles, shuffled: $3.9 \pm 0.04\%$, actual = 6.7% , $p = 0.047$).

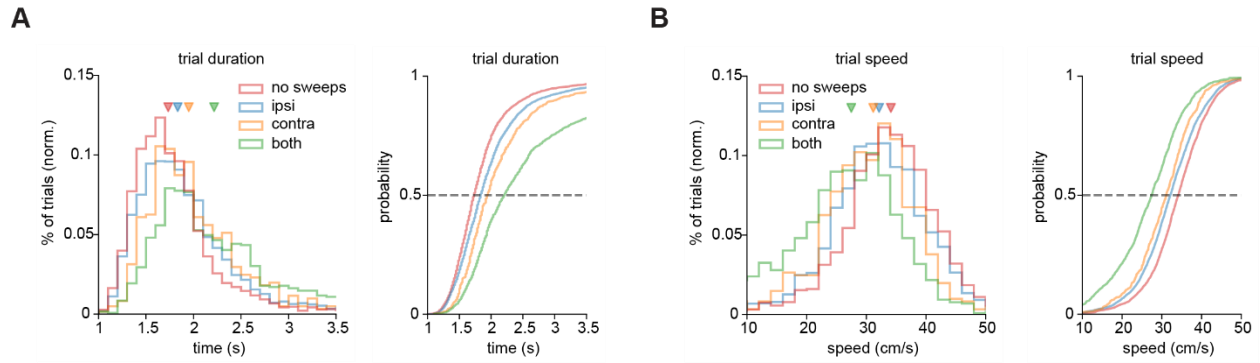


Figure S9: Effects of sweep presence and direction on trial duration and speed

(A) Left, trial durations for trials with no sweeps ($n = 2810$ trials, red), ipsi sweeps ($n = 3944$, blue), contra sweeps ($n = 607$, orange), and both ipsi and contra sweeps ($n = 1290$, green) pooled across sessions and animals ($n = 19$ sessions, 5 animals). Markers indicate medians of each group (1.73, 1.83, 1.95, and 2.22 s; IQRs: 1.52–2, 1.58–2.18, 1.68–2.35, 1.82–2.95 s, respectively, $p < 0.001$ for all comparisons, Mann-Whitney U tests, Bonferroni-corrected). Right, cumulative plot of each distribution in the left panel.

(B) Left, trial speeds for trials with no sweeps, ipsi sweeps, contra sweeps, and both ipsi and contra sweeps pooled across sessions and animals as in A. Speed was calculated as a single value across each trial as 2D distance traveled divided by trial duration. Markers indicate medians of each group (34.1, 32.1, 31.2, 27.5 cm/s; IQRs: 29.6–38.9, 27.3–37.2, 26.0–35.4, 21.3–32.7 cm/s, respectively, $p < 0.001$ for all comparisons, Mann-Whitney U tests, Bonferroni-corrected). Right, cumulative plot of each distribution in the left panel.